

# Gaussian Coefficients in Combinatorial Coefficient Calculus

**A  $q$ -binomial extension through Newton bases,  
Stirling–Eulerian transforms, Bell polynomials,  
series inversion, and asymptotic expansions**

A self-contained research companion to  
*Combinatorial Coefficient Calculus*

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## Abstract

Gaussian, or  $q$ -binomial, coefficients are more than weighted versions of ordinary binomial coefficients: they are the connection coefficients between monomials and factorial polynomials on a geometric grid. This observation places them directly inside the basis-change and coefficient-extraction framework of *Combinatorial Coefficient Calculus*. We develop that extension systematically. Finite products give convolution and inversion; affine changes of geometric nodes give explicit  $q$ -Stirling and colored-node formulae; weighted sorting gives  $q$ -Worpitzky and Stirling–Eulerian transforms. Ordinary Bell polynomials organize powers of  $q$ -products, parameter derivatives, local expansions at  $q = 1$  and at roots of unity, and closed coefficients for compositional inverses. Jackson differentiation is handled separately from ordinary composition, including a divided-difference chain rule that retains the necessary geometric shifts. Three asymptotic regimes are distinguished, and an application gives exact Gaussian extrapolation of moments of geometric uniform convolutions, including the Rvachev half-base case. All principal identities are proved; an accompanying exact-arithmetic program checks finite instances independently. The contribution is an integrated extension and a collection of explicit coefficient mechanisms, not a claim that the classical identities are new.

## Contents

|           |                                                                             |           |
|-----------|-----------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Scope, conventions, and the connection to the parent article</b>         | <b>3</b>  |
| <b>2</b>  | <b>Gaussian polynomials and their finite models</b>                         | <b>4</b>  |
| <b>3</b>  | <b>Finite products, convolutions, and Gaussian inversion</b>                | <b>5</b>  |
| <b>4</b>  | <b>Infinite coefficient series and terminating summations</b>               | <b>7</b>  |
| <b>5</b>  | <b>Newton bases: Gaussian coefficients as connection coefficients</b>       | <b>8</b>  |
| <b>6</b>  | <b><math>q</math>-Stirling coefficients and operator normal ordering</b>    | <b>10</b> |
| <b>7</b>  | <b><math>q</math>-Eulerian polynomials and the Stirling–Eulerian bridge</b> | <b>13</b> |
| <b>8</b>  | <b>Bell polynomials and the geometric-alphabet master formula</b>           | <b>14</b> |
| <b>9</b>  | <b>Jackson differentiation and geometric Taylor expansion</b>               | <b>16</b> |
| <b>10</b> | <b>An exact Jackson chain rule through divided differences</b>              | <b>17</b> |
| <b>11</b> | <b>Composition in a <math>q</math>-divided-power normalization</b>          | <b>18</b> |
| <b>12</b> | <b>Ordinary Lagrange inversion with Gaussian input</b>                      | <b>20</b> |
| <b>13</b> | <b>Parameter derivatives and weighted Gaussian sums</b>                     | <b>22</b> |
| <b>14</b> | <b>All Taylor coefficients at <math>q = 1</math></b>                        | <b>23</b> |
| <b>15</b> | <b>Roots of unity: polynomial evaluation and local jets</b>                 | <b>25</b> |
| <b>16</b> | <b>Three distinct asymptotic regimes</b>                                    | <b>26</b> |
| <b>17</b> | <b>Gaussian filters and geometric uniform convolutions</b>                  | <b>29</b> |
| <b>18</b> | <b>Verification, implementation, and further extensions</b>                 | <b>31</b> |
| <b>19</b> | <b>Conclusion</b>                                                           | <b>33</b> |
| <b>A</b>  | <b>A formula-selection guide</b>                                            | <b>34</b> |
|           | <b>References</b>                                                           | <b>35</b> |

# 1 Scope, conventions, and the connection to the parent article

The parent monograph [1] organizes combinatorics around polynomial basis changes, ordinary and exponential generating functions, Bell-polynomial composition, and Lagrange–Bürmann inversion. The present article extends those mechanisms rather than replacing the binomial sign in every formula by a Gaussian bracket. This distinction is essential: there are several inequivalent objects called  $q$ -Stirling,  $q$ -Bell, or  $q$ -Catalan numbers. The defining identities below, not the names, fix our conventions.

The parent source was consulted on the live main branch on 4 September 2026. The sections on formal series, Newton expansion, Bell polynomials, composition, Lagrange inversion, and Riordan arrays determine our points of contact. Since its chapter numbering can change, we refer to these topics rather than frozen page numbers. The statements here are not presented as Lean-checked results. The repository’s broader Fabius–Rvachev exposition already contains substantial  $q$ -calculus [2]; the classical ingredients repeated here serve as a self-contained foundation for the connections developed below.

## 1.1 Algebraic and analytic settings

For algebraic identities,  $q$  is initially an indeterminate. Polynomial identities live in  $\mathbb{Z}[q]$ ; divisions are performed in  $\mathbb{Q}(q)$  or in a specified extension. Formal power series in  $z$  are taken over a commutative  $\mathbb{Q}(q)$ -algebra. A product  $\prod_{j \geq 0} (1 - aq^j z)$  is not, without an additional topology or a coefficient definition, a product in  $\mathbb{Q}(q)[[z]]$ . We first define its coefficients algebraically or work analytically with  $|q| < 1$ , and only then use infinite-product notation.

The Gaussian polynomial is evaluated at  $q = 1$ ,  $q = 0$ , or a root of unity *after* polynomial cancellation. Expressions containing  $1/(q; q)_n$  or  $1/[n]_q!$  do not generally survive those specializations. Unless otherwise stated,  $n, k, N$  are nonnegative integers, empty products are one, impossible finite sums are zero, and  $0^0 = 1$  in coefficient formulae. An ordinary binomial coefficient with nonnegative upper index is zero outside its usual range. The default zero convention for Gaussian coefficients likewise concerns a *nonnegative* upper index; a separate negative-index convention appears in Section 4.

## 1.2 A map of the extension

| Parent mechanism                        | Gaussian extension in this article                                                                                         |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Factorial bases and Stirling inversion  | Geometric Newton bases; affine-geometric nodes; explicit $q$ -Stirling transforms (Sections 5 and 6).                      |
| Eulerian and Worpitzky identities       | Descent–major-index enumeration and the Gaussian Stirling–Eulerian bridge (Section 7).                                     |
| Bell polynomials and coefficient powers | Geometric power sums, signed alphabets, and $q$ -normalized composition (Sections 8 and 11).                               |
| Faà di Bruno calculus                   | Jackson product rules and a divided-difference chain rule on geometric nodes (Sections 9 and 10).                          |
| Lagrange and Riordan inversion          | Closed inverse coefficients for finite $q$ -products, and Gaussian Pascal arrays (Sections 11 and 12).                     |
| Bernoulli and asymptotic calculus       | All-order local $q$ -expansions, root-of-unity regularization, and a uniform double-scaling expansion (Sections 13 to 16). |

Three distinctions will recur. A change from  $n!$  to  $[n]_q!$  is a *normalization of coefficients*; replacing arithmetic nodes by geometric nodes is a *change of polynomial basis*; replacing the ordinary derivative by  $D_q$  is a *change of operator*. They interact, but they are not the same construction.

## 2 Gaussian polynomials and their finite models

### 2.1 Definitions, Pascal identities, and symmetry

Set

$$[n]_q = 1 + q + \cdots + q^{n-1}, \quad [n]_q! = \prod_{j=1}^n [j]_q, \quad (a; q)_N = \prod_{j=0}^{N-1} (1 - aq^j). \quad (2.1)$$

Thus  $[0]_q = 0$ ,  $[0]_q! = 1$ , and  $(q; q)_n = (1 - q)^n [n]_q!$ . The notation in this section is standard; see [7] for the classical  $q$ -calculus conventions.

**Definition 2.1.** For  $0 \leq k \leq n$ , define

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \prod_{i=1}^k \frac{1 - q^{n-k+i}}{1 - q^i}. \quad (2.2)$$

For  $k < 0$  or  $k > n \geq 0$ , set  $G_{n,k}(q) = 0$ .

**Theorem 2.2** (Pascal identities and polynomiality). *The quotients in Equation (2.2) are polynomials in  $\mathbb{Z}[q]$ . For  $n \geq 1$  they satisfy*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \quad (2.3)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q. \quad (2.4)$$

For  $0 \leq k \leq n$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q, \quad G_{n,k}(q^{-1}) = q^{-k(n-k)} G_{n,k}(q), \quad G_{n,k}(1) = \binom{n}{k}. \quad (2.5)$$

*Their coefficients are nonnegative integers, and their degree is  $k(n-k)$ .*

*Proof.* For  $0 < k < n$ , put the two terms in Equation (2.3) over the common denominator  $(q; q)_k (q; q)_{n-k}$ . Their numerator is  $(q; q)_{n-1} \{1 - q^k + q^k(1 - q^{n-k})\} = (q; q)_n$ . The boundary cases are immediate. This proves polynomiality and nonnegativity inductively, starting with  $G_{0,0} = 1$ . Symmetry follows from the factorial quotient, and gives Equation (2.4). Replacing each factor  $1 - q^{-j}$  by  $-q^{-j}(1 - q^j)$  gives the reciprocity law. The factors  $(1 - q^a)/(1 - q)$  tend to  $a$ , giving the ordinary binomial limit. The degree assertion follows from the product, or from the weighted model in the next proof, whose minimum and maximum weights are both attained.  $\square$

### 2.2 Subsets, rectangle partitions, and words

**Theorem 2.3** (Three polynomial models). *For  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{0 \leq i_1 < \cdots < i_k < n} q^{i_1 + \cdots + i_k - \binom{k}{2}}, \quad (2.6)$$

$$= \sum_{0 \leq b_1 \leq \cdots \leq b_k \leq n-k} q^{b_1 + \cdots + b_k}. \quad (2.7)$$

*Equivalently, it enumerates partitions fitting in a  $k \times (n-k)$  rectangle by their size. It also enumerates binary words with  $k$  ones and  $n-k$  zeros by the number of pairs consisting of a zero to the left of a one.*

*Proof.* Let the first sum be  $H_{n,k}$ . Subsets not containing  $n-1$  contribute  $H_{n-1,k}$ ; deleting  $n-1$  from the others decreases the exponent by  $(n-1) - (k-1) = n-k$ . Thus  $H$  satisfies Equation (2.4) with the same boundary values as  $G$ . The substitution  $b_j = i_j - (j-1)$  gives the second sum. Reversing the  $b_j$  gives a partition inside the rectangle. In the associated binary word, the  $j$ th one has exactly  $i_j - (j-1)$  zeros before it. Summing gives the stated word statistic.  $\square$

Reversal of the word changes that statistic to  $k(n-k)$  minus itself, which explains reciprocity without division. For example,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + q^3 + q^4. \quad (2.8)$$

The two objects of weight 2 are the rectangle partitions  $(2, 0)$  and  $(1, 1)$ .

### 2.3 Multinomials, finite fields, and flags

For  $n_1 + \cdots + n_r = n$ , write

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q = \frac{[n]_q!}{\prod_{i=1}^r [n_i]_q!} = \prod_{i=1}^r \begin{bmatrix} n - \sum_{j<i} n_j \\ n_i \end{bmatrix}_q. \quad (2.9)$$

The right side proves polynomiality. Assigning the positions of letter 1, then letter 2, and so on shows that it enumerates words with these multiplicities by the usual inversion statistic; reversal also gives the complementary convention.

**Proposition 2.4** (Subspace and flag counts). *If  $Q$  is a prime power,  $G_{n,k}(Q)$  counts the  $k$ -dimensional subspaces of  $\mathbb{F}_Q^n$ . The multinomial in Equation (2.9), evaluated at  $q = Q$ , counts flags whose successive dimension increments are  $n_1, \dots, n_r$ .*

*Proof.* There are  $(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1})$  ordered independent  $k$ -tuples in  $\mathbb{F}_Q^n$ . Each  $k$ -subspace has  $(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$  ordered bases. Dividing and canceling the powers of  $Q$  gives Equation (2.2). Choose each next subspace in the quotient by the preceding one to obtain the flag product.  $\square$

The indeterminate  $q$  is not itself a field size. The finite-field interpretation applies only at prime powers; the polynomial identities hold independently of it.

## 3 Finite products, convolutions, and Gaussian inversion

### 3.1 The two finite coefficient theorems

**Theorem 3.1** (Finite  $q$ -binomial theorem). *For  $N \geq 0$ ,*

$$(z; q)_N = \sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (3.1)$$

*For  $N \geq 1$ , its reciprocal has the formal expansion*

$$\frac{1}{(z; q)_N} = \sum_{m \geq 0} \begin{bmatrix} N + m - 1 \\ m \end{bmatrix}_q z^m. \quad (3.2)$$

*Proof.* In the product defining  $(z; q)_N$ , choose the term  $-zq^j$  at a subset of  $k$  positions; Equation (2.6) proves Equation (3.1). In the reciprocal product choose a nonnegative multiplicity of each  $q^j z$ . Its degree- $m$  coefficient is

$$\sum_{0 \leq i_1 \leq \dots \leq i_m < N} q^{i_1 + \dots + i_m}.$$

The substitution  $b_j = i_j + j - 1$  gives a strictly increasing  $m$ -subset of  $\{0, \dots, N + m - 2\}$ , with the shift  $\binom{m}{2}$  removed from its weight. This is Equation (2.6) with upper index  $N + m - 1$ .  $\square$

### 3.2 Vandermonde identities with all phases visible

**Theorem 3.2** (Two-block and multiblock convolution). *For  $m, n, k \geq 0$ ,*

$$\begin{bmatrix} m+n \\ k \end{bmatrix}_q = \sum_{j=0}^k q^{(m-j)(k-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ k-j \end{bmatrix}_q. \quad (3.3)$$

*Terms with invalid indices are omitted, so the displayed powers are nonnegative on the contributing range. More generally, if  $N = N_1 + \dots + N_r$ , then*

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \sum_{\substack{k_1 + \dots + k_r = k \\ 0 \leq k_i \leq N_i}} q^{\sum_{i < j} (N_i - k_i)k_j} \prod_{i=1}^r \begin{bmatrix} N_i \\ k_i \end{bmatrix}_q. \quad (3.4)$$

*For  $m, n \geq 1$ , a reciprocal-product version is*

$$\begin{bmatrix} m+n+k-1 \\ k \end{bmatrix}_q = \sum_{j=0}^k q^{m(k-j)} \begin{bmatrix} m+j-1 \\ j \end{bmatrix}_q \begin{bmatrix} n+k-j-1 \\ k-j \end{bmatrix}_q. \quad (3.5)$$

*Proof.* Split the finite product as  $(z; q)_{m+n} = (z; q)_m (q^m z; q)_n$  and compare  $z^k$  in Equation (3.1). The exponent left after canceling  $q^{\binom{k}{2}}$  is  $\binom{j}{2} + \binom{k-j}{2} + m(k-j) - \binom{k}{2} = (m-j)(k-j)$ . For several blocks the same subtraction is

$$\sum_i \binom{k_i}{2} + \sum_i k_i \sum_{h < i} N_h - \binom{k}{2} = \sum_{h < i} (N_h - k_h) k_i.$$

Finally split the reciprocal product and use Equation (3.2). This time no triangular powers are canceled; the second block contributes exactly  $q^{m(k-j)}$ .  $\square$

The phase in Equation (3.3) is not optional. It records crossings between two blocks, or, in the binary model, zeros in the first block lying before ones in the second.

### 3.3 Annihilation and inversion

**Corollary 3.3** (Gaussian annihilator). *For an indeterminate  $x$ ,*

$$\sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q x^j = (x; q)_N. \quad (3.6)$$

*In particular the sum at  $x = 1$  is zero for  $N > 0$ . If  $q \neq 0$  and  $x = q^{-r}$  with  $0 \leq r < N$ , it is again zero.*

*Proof.* This is Equation (3.1); in the two specializations one factor of the product is zero.  $\square$

**Theorem 3.4** (Gaussian binomial inversion). *For any two sequences with values in a  $\mathbb{Z}[q]$ -module,*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k. \quad (3.7)$$

*Proof.* Substitute the first relation in the right side. The coefficient of  $a_j$  becomes, using factorial cancellation,

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put  $r = n - k$  and apply Theorem 3.3. The inner sum is zero unless  $j = n$ , when it is one. Both matrices are unit lower triangular, so this left inverse is also the inverse; alternatively, the same finite coefficient computation gives the reverse implication.  $\square$

**Proposition 3.5** (Möbius inversion on subspaces). *If  $U \subseteq V$  are finite-dimensional vector spaces over  $\mathbb{F}_Q$  and  $d = \dim(V/U)$ , the Möbius function of the subspace lattice is*

$$\mu(U, V) = (-1)^d Q^{\binom{d}{2}}. \quad (3.8)$$

Consequently, summation over subspaces of a fixed dimension gives Equation (3.7) at  $q = Q$ .

*Proof.* The defining relation for  $\mu$  is  $\sum_{U \subseteq W \subseteq V} \mu(U, W) = \delta_{U,V}$ . There are  $G_{d,j}(Q)$  intermediate subspaces with  $\dim(W/U) = j$ . The proposed formula satisfies the defining relation by Theorem 3.3, and the relation determines  $\mu$  uniquely by increasing interval length.  $\square$

## 4 Infinite coefficient series and terminating summations

The finite theory is enough for most of this article. The following infinite series give a convenient algebraic language and a route to terminating hypergeometric summations; compare [7, 8, 9].

**Theorem 4.1** (The parameter  $q$ -binomial series). *In  $\mathbb{Q}(q, a)[[z]]$  define*

$$H_a(z) = \sum_{m \geq 0} \frac{(a; q)_m}{(q; q)_m} z^m. \quad (4.1)$$

*It is the unique series with constant term one satisfying*

$$(1 - z)H_a(z) = (1 - az)H_a(qz). \quad (4.2)$$

*When  $|q| < 1$  and  $|z| < 1$ ,*

$$H_a(z) = \frac{(az; q)_\infty}{(z; q)_\infty}. \quad (4.3)$$

*Moreover,*

$$\sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q (a; q)_j (b; q)_{n-j} a^{n-j} = (ab; q)_n. \quad (4.4)$$

*Proof.* Writing  $H_a = \sum h_m z^m$  in Equation (4.2) gives  $(1 - q^m)h_m = (1 - aq^{m-1})h_{m-1}$  for  $m \geq 1$ , which is solved by the finite product in Equation (4.1). This proves existence and uniqueness formally. Analytically the products converge uniformly on compact  $z$ -sets away from denominator zeros. Their ratio has constant term one and satisfies the same equation, proving Equation (4.3). The product  $H_a(z)H_b(az)$  satisfies Equation (4.2) with  $a$  replaced by  $ab$ : the two intermediate factors  $1 - az$  cancel. Uniqueness therefore gives  $H_a(z)H_b(az) = H_{ab}(z)$ . Compare the coefficient of  $z^n$  and multiply by  $(q; q)_n$ .  $\square$

In particular, define the *normalized  $q$ -exponentials* by

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{[n]_q!}, \quad E_q(z) = \sum_{n \geq 0} \frac{q^{\binom{n}{2}} z^n}{[n]_q!}. \quad (4.5)$$

The normalization differs from the common convention with  $(q; q)_n$  in the denominator by the substitution  $z \mapsto (1 - q)z$ . For  $|q| < 1$ ,

$$e_q(z) = \frac{1}{((1 - q)z; q)_\infty}, \quad E_q(z) = (- (1 - q)z; q)_\infty, \quad e_q(z)E_q(-z) = 1. \quad (4.6)$$

The first identity follows from Theorem 4.1. For the second, expand a finite product and pass coefficientwise to the limit, using  $G_{N,k}(q) \rightarrow 1/(q; q)_k$ . The inverse relation also follows formally from Theorem 3.3. The analytic series for  $e_q$  is used inside its disk  $|1 - q| < 1$ ;  $E_q$  is entire in  $z$ .

**Theorem 4.2** (The two terminating  $q$ -Chu–Vandermonde sums). *For  $n \geq 0$ , as rational identities in generic parameters,*

$$\sum_{j=0}^n \frac{(q^{-n}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} \left( \frac{cq^n}{a} \right)^j = \frac{(c/a; q)_n}{(c; q)_n}, \quad (4.7)$$

$$\sum_{j=0}^n \frac{(q^{-n}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} q^j = a^n \frac{(c/a; q)_n}{(c; q)_n}. \quad (4.8)$$

*Numerical specialization requires nonzero displayed denominators, or prior rational cancellation.*

*Proof.* Divide Equation (4.4) by  $a^n(b; q)_n$  and use

$$\left[ \begin{matrix} n \\ j \end{matrix} \right]_q \frac{(b; q)_{n-j}}{(b; q)_n} = \frac{(q^{-n}; q)_j}{(q; q)_j (q^{1-n}/b; q)_j} \left( \frac{q}{b} \right)^j. \quad (4.9)$$

This equality follows by reversing the  $j$  factors canceled from each product. Set  $c = q^{1-n}/b$ . Reversing the  $n$  factors on the right side also gives  $(ab; q)_n / (a^n(b; q)_n) = (c/a; q)_n / (c; q)_n$ , proving Equation (4.7). In that identity replace  $(q, a, c)$  by  $(q^{-1}, a^{-1}, c^{-1})$  and use

$$(x; q^{-1})_j = (-x)^j q^{-\binom{j}{2}} (x^{-1}; q)_j.$$

The transformed summand acquires the net factor  $q^j$ , and the transformed right side acquires  $a^n$ . This proves Equation (4.8).  $\square$

**Proposition 4.3** (A negative upper index). *For  $m \geq 1$  and  $k \geq 0$ , define  $\left[ \begin{matrix} -m \\ k \end{matrix} \right]_q = \prod_{j=0}^{k-1} (1 - q^{-m-j}) / (q; q)_k$ . Then*

$$\left[ \begin{matrix} -m \\ k \end{matrix} \right]_q = (-1)^k q^{-mk - \binom{k}{2}} \left[ \begin{matrix} m + k - 1 \\ k \end{matrix} \right]_q. \quad (4.10)$$

*Proof.* Factor  $-q^{-m-j}$  from the  $j$ th numerator factor and reverse the remaining factors. Their quotient by  $(q; q)_k$  is the Gaussian polynomial on the right.  $\square$

This limited extension has an unambiguous nonnegative lower index. Extending *both* indices to all integers requires further conventions; see the primary treatment [6].

## 5 Newton bases: Gaussian coefficients as connection coefficients

### 5.1 The general node calculus

Let  $a_0, a_1, \dots$  be elements of a commutative ring and set

$$P_0^a(x) = 1, \quad P_k^a(x) = \prod_{j=0}^{k-1} (x - a_j). \quad (5.1)$$



Since these polynomials are monic of successive degrees, there are unique arrays  $S_a$  and  $s_a$  such that

$$x^n = \sum_{k=0}^n S_a(n, k) P_k^a(x), \quad P_n^a(x) = \sum_{k=0}^n s_a(n, k) x^k. \quad (5.2)$$

This common framework for binomial, Stirling, and Gaussian coefficients is classical; Voigt's short paper [3] is particularly close in spirit to the basis-change program of the parent monograph.

For a finite list  $A = (a_0, \dots, a_k)$ , the elementary and complete homogeneous polynomials are defined by

$$e_m(A) = \sum_{i_1 < \dots < i_m} a_{i_1} \cdots a_{i_m}, \quad h_m(A) = \sum_{i_1 \leq \dots \leq i_m} a_{i_1} \cdots a_{i_m}.$$

Both have value one at  $m = 0$ ;  $e_m = 0$  above the list length.

**Theorem 5.1** (Universal finite node formulae). *For  $0 \leq k \leq n$ ,*

$$S_a(n, k) = h_{n-k}(a_0, \dots, a_k), \quad (5.3)$$

$$s_a(n, k) = (-1)^{n-k} e_{n-k}(a_0, \dots, a_{n-1}). \quad (5.4)$$

*The arrays are inverse lower-triangular matrices. In particular,*

$$\sum_{j=k}^n s_a(n, j) S_a(j, k) = \delta_{n,k}. \quad (5.5)$$

*Their recurrences are*

$$S_a(n+1, k) = S_a(n, k-1) + a_k S_a(n, k), \quad (5.6)$$

$$s_a(n+1, k) = s_a(n, k-1) - a_n s_a(n, k). \quad (5.7)$$

*Proof.* The identity  $xP_k = P_{k+1} + a_k P_k$  gives Equation (5.6). Separating monomials in  $h_m(a_0, \dots, a_k)$  according as they contain  $a_k$  gives the same recurrence for Equation (5.3), including the boundary values. Formula Equation (5.4) follows by expanding the product, and multiplication by  $x - a_n$  gives Equation (5.7). Substituting one basis expansion into the other proves matrix inversion by uniqueness of monomial coefficients.  $\square$

All these identities are valid even when nodes coincide. Only the next representation requires distinct nodes.

**Theorem 5.2** (Divided differences and a nonrecursive spectral sum). *Over a field, assume  $a_0, \dots, a_k$  are distinct. Then*

$$S_a(n, k) = \sum_{j=0}^k \frac{a_j^n}{\prod_{0 \leq i \leq k, i \neq j} (a_j - a_i)}. \quad (5.8)$$

*More generally, the coefficient of  $P_k^a$  in the Newton interpolant to values  $f(a_0), \dots, f(a_M)$  is*

$$f[a_0, \dots, a_k] = \sum_{j=0}^k \frac{f(a_j)}{\prod_{i \neq j} (a_j - a_i)}. \quad (5.9)$$

*Thus its degree- $M$  interpolating polynomial is  $\sum_{k=0}^M f[a_0, \dots, a_k] P_k^a(x)$ .*

*Proof.* The Lagrange interpolant on the first  $k+1$  nodes is

$$L_k(x) = \sum_{j=0}^k f(a_j) \frac{\prod_{i \neq j} (x - a_i)}{\prod_{i \neq j} (a_j - a_i)}.$$

Its leading coefficient is the sum in Equation (5.9). In a Newton expansion, every term with index greater than  $k$  vanishes on these nodes, and every term with index less than  $k$  has smaller degree. Hence that leading coefficient is precisely the  $k$ th Newton coefficient. Apply this to  $f(x) = x^n$  for Equation (5.8).  $\square$

## 5.2 Geometric nodes and the Gaussian basis pair

**Theorem 5.3** (The Gaussian Newton pair). *Put  $P_k(x) = \prod_{j=0}^{k-1} (x - q^j)$ . Then*

$$x^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q P_k(x), \quad (5.10)$$

$$P_n(x) = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q x^k. \quad (5.11)$$

*Proof.* By Equation (5.3), the first coefficient is  $h_{n-k}(1, q, \dots, q^k)$ , which is  $G_{n,k}(q)$  by Equation (3.2). The second is the elementary symmetric coefficient in the same geometric list, and Equation (3.1) gives exactly the stated phase. Equivalently, the first recurrence is Equation (2.3) with  $a_k = q^k$ .  $\square$

This is the precise sense in which Gaussian coefficients are Stirling-type coefficients: the nodes  $0, 1, 2, \dots$  produce ordinary Stirling numbers, while  $1, q, q^2, \dots$  produce Gaussian polynomials. At  $q = 1$  these geometric nodes all equal one, so Equation (5.10) becomes the ordinary binomial expansion  $x^n = \sum_k \binom{n}{k} (x-1)^k$ , not the ordinary Stirling expansion.

For generic nonzero  $q$ , the explicit geometric-grid formula is

$$f[1, q, \dots, q^k] = \frac{q^{-\binom{k}{2}}}{(q; q)_k} \sum_{j=0}^k (-1)^j q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q f(q^j). \quad (5.12)$$

Indeed,

$$\prod_{i \neq j} (q^j - q^i) = (-1)^j q^{j^k - j(j+1)/2} (q; q)_j (q; q)_{k-j}, \quad (5.13)$$

obtained by separating  $i < j$  from  $i > j$ . Substitution in Equation (5.9) proves Equation (5.12). This is a finite interpolation formula; an infinite Newton expansion would need separate convergence assumptions.

## 5.3 An affine-geometric master connection

**Theorem 5.4** (Affine-geometric nodes). *For nodes  $a_j = A + Bq^j$ , the connection coefficients are*

$$S_{A+Bq^\bullet}(n, k) = \sum_{d=k}^n \binom{n}{d} A^{n-d} B^{d-k} \begin{bmatrix} d \\ k \end{bmatrix}_q. \quad (5.14)$$

*This is a polynomial identity in  $A, B, q$ , including  $B = 0$ .*

*Proof.* First take  $B$  invertible and write  $x = A + By$ . Then  $P_k^a(x) = B^k P_k(y)$ . Expand  $(A + By)^n$  ordinarily, then expand each  $y^d$  by Equation (5.10). The coefficient of  $B^k P_k(y)$  is Equation (5.14). Both sides are polynomials, so the identity descends from  $\mathbb{Q}(A, B, q)$  to  $\mathbb{Z}[A, B, q]$  and can be specialized at  $B = 0$ .  $\square$

This theorem is a compact engine for several families that otherwise look unrelated. The next section makes its  $q$ -arithmetic specializations explicit.

# 6 $q$ -Stirling coefficients and operator normal ordering

## 6.1 The convention and three closed formulae

Define

$$F_k(x) = \prod_{j=0}^{k-1} (x - [j]_q), \quad x^n = \sum_{k=0}^n S_q(n, k) F_k(x), \quad F_n(x) = \sum_{k=0}^n s_q(n, k) x^k. \quad (6.1)$$

Thus

$$S_q(n+1, k) = S_q(n, k-1) + [k]_q S_q(n, k). \quad (6.2)$$

This is the convention used for the type- $A$  array in [4]. The rescaled array  $q^{\binom{k}{2}} S_q(n, k)$  is also frequently called a  $q$ -Stirling array; it must not be silently substituted for  $S_q$ .

**Theorem 6.1** (Explicit Gaussian formulae for  $q$ -Stirling arrays). *For  $0 \leq k \leq n$ ,*

$$S_q(n, k) = h_{n-k}([0]_q, [1]_q, \dots, [k]_q), \quad (6.3)$$

$$S_q(n, k) = (1-q)^{k-n} \sum_{d=k}^n (-1)^{d-k} \binom{n}{d} \left[ \begin{matrix} d \\ k \end{matrix} \right]_q, \quad (6.4)$$

$$S_q(n, k) = \frac{q^{-\binom{k}{2}}}{[k]_q!} \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \left[ \begin{matrix} k \\ j \end{matrix} \right]_q [j]_q^n. \quad (6.5)$$

The signed inverse coefficients satisfy

$$s_q(n, k) = (-1)^{n-k} e_{n-k}([0]_q, \dots, [n-1]_q), \quad (6.6)$$

$$s_q(n, k) = (1-q)^{k-n} \sum_{d=k}^n (-1)^{d-k} \binom{d}{k} q^{\binom{n-d}{2}} \left[ \begin{matrix} n \\ d \end{matrix} \right]_q. \quad (6.7)$$

All apparent singularities in Equations (6.4) and (6.7) at  $q = 1$  are removable. Formula Equation (6.5) is initially for generic nonzero  $q$ ; its rational expression equals the same polynomial.

*Proof.* The symmetric-function formulae are Theorem 5.1. Since  $[j]_q = (1-q)^{-1} - (1-q)^{-1}q^j$ , substitution in Equation (5.14) proves Equation (6.4). For the spectral formula, the pairwise differences are

$$[j]_q - [i]_q = \begin{cases} q^i[j-i]_q, & i < j, \\ -q^j[i-j]_q, & i > j. \end{cases}$$

Therefore

$$\prod_{i \neq j} ([j]_q - [i]_q) = (-1)^{k-j} q^{jk-j(j+1)/2} [j]_q! [k-j]_q!.$$

Use Equation (5.8) and the identity  $jk - j(j+1)/2 = \binom{k}{2} - \binom{k-j}{2}$ . For the final formula put  $y = 1 - (1-q)x$ , so that  $F_n(x) = (-1)^n (1-q)^{-n} P_n(y)$ . Insert Equation (5.11), expand  $y^d$  ordinarily, and compare  $x^k$ . Polynomiality from the definitions proves the cancellation claims.  $\square$

The first few rows are

| $n \backslash k$ | 0 | 1 | 2          | 3          | 4 |
|------------------|---|---|------------|------------|---|
| 0                | 1 |   |            |            |   |
| 1                | 0 | 1 |            |            |   |
| 2                | 0 | 1 | 1          |            |   |
| 3                | 0 | 1 | $2+q$      | 1          |   |
| 4                | 0 | 1 | $3+3q+q^2$ | $3+2q+q^2$ | 1 |

(6.8)

In particular  $S_q(n, 1) = 1$  for  $n \geq 1$ . The limit  $q \rightarrow 1$  recovers ordinary Stirling numbers, while  $q = 0$  gives  $S_0(n, k) = \binom{n-1}{k-1}$  for  $1 \leq k \leq n$ : all positive-index nodes have become one, and Equation (6.3) counts weak compositions of  $n-k$  into  $k$  positions.

## 6.2 Colored and type- $B$ node specializations

**Corollary 6.2** (Nodes  $[r + mj]_q$ ). Fix integers  $m \geq 1$  and  $r \geq 0$ , and define  $S_q^{(m,r)}(n, k)$  using the nodes  $a_j = [r + mj]_q$ . Then

$$S_q^{(m,r)}(n, k) = (1 - q)^{k-n} \sum_{d=k}^n \binom{n}{d} (-q^r)^{d-k} \left[ \begin{matrix} d \\ k \end{matrix} \right]_{q^m}. \quad (6.9)$$

It is a polynomial with nonnegative integer coefficients. For  $(m, r) = (2, 1)$  its defining recurrence is

$$S_q^{(2,1)}(n+1, k) = S_q^{(2,1)}(n, k-1) + [2k+1]_q S_q^{(2,1)}(n, k), \quad (6.10)$$

the type- $B$  convention discussed in [4].

*Proof.* Write  $[r + mj]_q = (1 - q)^{-1} - q^r(1 - q)^{-1}(q^m)^j$  and use Theorem 5.4. Nonnegativity follows from the complete homogeneous formula Equation (5.3); Equation (6.10) is Equation (5.6).  $\square$

No signed-permutation interpretation is needed to prove Equation (6.9). It gives a finite Gaussian formula for the entire node family, including its boundary values  $S_q^{(m,r)}(n, 0) = [r]_q^n$ .

## 6.3 The $q$ -Weyl relation and a $q$ -Dobinski identity

Let  $(Xf)(x) = xf(x)$  and

$$D_q f(x) = \frac{f(x) - f(qx)}{(1 - q)x}, \quad D_q x^n = [n]_q x^{n-1}. \quad (6.11)$$

On polynomials this is defined by the second equality even at  $x = 0$ . A direct calculation gives

$$D_q X - qX D_q = I. \quad (6.12)$$

**Theorem 6.3** (Normal ordering). Put  $C_q(n, k) = q^{\binom{k}{2}} S_q(n, k)$ . Then, as polynomial operators,

$$(X D_q)^n = \sum_{k=0}^n C_q(n, k) X^k D_q^k. \quad (6.13)$$

The polynomial  $T_{n,q}(z) = \sum_k C_q(n, k) z^k$  has the identity

$$T_{n,q}(z) = \frac{1}{e_q(z)} \sum_{j \geq 0} [j]_q^n \frac{z^j}{[j]_q!}. \quad (6.14)$$

This holds formally in  $z$  over  $\mathbb{Q}(q)$ ; for  $0 < q < 1$  it also holds analytically when  $|(1 - q)z| < 1$ .

*Proof.* The relation Equation (6.12) implies  $D_q X^k = [k]_q X^{k-1} + q^k X^k D_q$  by induction on  $k$ . Multiplying the proposed normal form by  $X D_q$  gives the coefficient recurrence  $C_q(n+1, k) = [k]_q C_q(n, k) + q^{k-1} C_q(n, k-1)$ . This is exactly Equation (6.2) after the stated rescaling, and the initial operators agree. For Equation (6.14), evaluate Equation (6.1) at  $x = [j]_q$ . When  $j \geq k$ ,

$$F_k([j]_q) = q^{\binom{k}{2}} \frac{[j]_q!}{[j-k]_q!},$$

and the value is zero when  $j < k$ . Multiply by  $z^j/[j]_q!$ , sum over  $j$ , and interchange the finite  $k$ -sum with the formal series. The result is  $e_q(z) T_{n,q}(z)$ . Analytically,  $[j]_q$  is bounded for  $0 < q < 1$ , so the same sums converge in the stated disk.  $\square$

The normalization in Equation (6.14) is therefore forced by the  $q$ -Weyl operator, not chosen by analogy with the ordinary Dobinski formula.

## 7 $q$ -Eulerian polynomials and the Stirling–Eulerian bridge

For a permutation  $\pi = (\pi_1, \dots, \pi_n)$ , let  $\text{Des}(\pi) = \{i : \pi_i > \pi_{i+1}\}$ ,  $\text{des}(\pi) = |\text{Des}(\pi)|$ , and  $\text{maj}(\pi) = \sum_{i \in \text{Des}(\pi)} i$ . Define

$$A_n(t, q) = \sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} q^{\text{maj}(\pi)} = \sum_{j=0}^{n-1} A_{n,j}(q) t^j \quad (n \geq 1). \quad (7.1)$$

The major index, rather than an unspecified inversion statistic, is important in the identities below.

**Theorem 7.1** ( $q$ -Worpitzky and a finite Eulerian sum). *For  $n \geq 1$ ,*

$$\sum_{m \geq 0} [m+1]_q^n t^m = \frac{A_n(t, q)}{(t; q)_{n+1}}. \quad (7.2)$$

Consequently, for  $m \geq 0$ ,

$$[m+1]_q^n = \sum_{j=0}^{n-1} A_{n,j}(q) \begin{bmatrix} m+n-j \\ n \end{bmatrix}_q, \quad (7.3)$$

$$A_{n,j}(q) = \sum_{r=0}^j (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n+1 \\ r \end{bmatrix}_q [j-r+1]_q^n. \quad (7.4)$$

*Proof.* Expand  $[m+1]_q^n$  as the weight sum of vectors  $(a_1, \dots, a_n) \in \{0, \dots, m\}^n$ , with weight  $q^{\sum a_i}$ . Sort the labels by decreasing  $a_i$ , resolving equal entries by increasing label. This gives a unique permutation  $\pi$  such that  $a_{\pi_1} \geq \dots \geq a_{\pi_n}$ , with a strict inequality at every descent of  $\pi$ . Put  $d = \text{des}(\pi)$  and subtract from the  $i$ th sorted entry the number of descents at positions at least  $i$ . The resulting sequence is  $m-d \geq b_1 \geq \dots \geq b_n \geq 0$ , and its weight is shifted by  $\text{maj}(\pi)$ . Conversely, this construction reconstructs exactly the vectors producing  $\pi$ . Write  $e_0 = m-d-b_1$ ,  $e_i = b_i - b_{i+1}$  for  $1 \leq i < n$ , and  $e_n = b_n$ . Then  $m = d + \sum_{i=0}^n e_i$  and  $\sum b_i = \sum_{i=1}^n i e_i$ . The generating function for these gaps is

$$t^d q^{\text{maj}(\pi)} \prod_{i=0}^n \frac{1}{1 - t q^i}.$$

Sum over permutations to get Equation (7.2). Extract  $t^m$  using Equation (3.2) for Equation (7.3); multiply by  $(t; q)_{n+1}$  and use Equation (3.1) for Equation (7.4).  $\square$

**Theorem 7.2** (Gaussian Stirling–Eulerian transforms). *For  $1 \leq k \leq n$ ,*

$$q^{\binom{k}{2}} [k]_q! S_q(n, k) = \sum_{\ell=1}^k q^{k(k-\ell)} A_{n, \ell-1}(q) \begin{bmatrix} n-\ell \\ k-\ell \end{bmatrix}_q. \quad (7.5)$$

Conversely,

$$A_n(t, q) = \sum_{k=1}^n q^{\binom{k}{2}} [k]_q! S_q(n, k) t^{k-1} (t q^{k+1}; q)_{n-k}. \quad (7.6)$$

*Proof.* Set  $U_k = q^{\binom{k}{2}} [k]_q! S_q(n, k)$ . The Newton expansion at  $x = [m+1]_q$  gives  $[m+1]_q^n = \sum_{k=1}^n U_k G_{m+1,k}(q)$ . Since

$$\sum_{m \geq 0} G_{m+1,k}(q) t^m = \frac{t^{k-1}}{(t; q)_{k+1}},$$

comparison with Equation (7.2) proves Equation (7.6). To invert it, we first establish the finite identity

$$\sum_{r=0}^N \begin{bmatrix} N \\ r \end{bmatrix}_q q^{r(r-1)} u^r (uq^r; q)_{N-r} = 1. \quad (7.7)$$

The coefficient of  $u^j$  on the left equals

$$\begin{bmatrix} N \\ j \end{bmatrix}_q q^{\binom{j}{2}} \sum_{r=0}^j (-1)^{j-r} q^{\binom{r}{2}} \begin{bmatrix} j \\ r \end{bmatrix}_q,$$

which is  $\delta_{j,0}$  by Theorem 3.3. Set  $N = n - \ell$  and  $u = tq^{\ell+1}$  in Equation (7.7) and multiply by  $t^{\ell-1}$ . This expresses  $t^{\ell-1}$  as

$$\sum_{k=\ell}^n q^{k(k-\ell)} \begin{bmatrix} n-\ell \\ k-\ell \end{bmatrix}_q t^{k-1} (tq^{k+1}; q)_{n-k}.$$

Insert this expansion for each monomial of  $A_n(t, q)$  in Equation (7.6). These basis polynomials have distinct lowest powers, so their coefficients are unique. The resulting coefficient of the  $k$ th basis polynomial is Equation (7.5).  $\square$

Identity Equation (7.5) is classical; its normalization and history are recorded in [4]. The proof here exhibits it as another triangular coefficient transform, directly parallel to the ordinary Stirling–Eulerian relations in the parent monograph. For example  $A_3(t, q) = 1 + (2q + 2q^2)t + q^3t^2$ , and at  $(n, k) = (3, 2)$  both sides of Equation (7.5) equal  $2q + 3q^2 + q^3$ .

## 8 Bell polynomials and the geometric-alphabet master formula

### 8.1 The ordinary Bell engine remains ordinary

We use the complete exponential Bell polynomial

$$\mathcal{Y}_n(x_1, \dots, x_n) = n! \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum j m_j = n}} \prod_{j=1}^n \frac{1}{m_j!} \left( \frac{x_j}{j!} \right)^{m_j}, \quad \mathcal{Y}_0 = 1. \quad (8.1)$$

Its generating identity is

$$\exp \left( \sum_{j \geq 1} x_j \frac{z^j}{j!} \right) = \sum_{n \geq 0} \mathcal{Y}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (8.2)$$

To prove it, expand each exponential  $\exp(x_j z^j / j!)$  and multiply; the coefficient condition is exactly  $\sum j m_j = n$ . The partial Bell polynomial  $\mathcal{B}_{n,k}$  is the same sum with the additional condition  $\sum m_j = k$ . Thus  $\mathcal{Y}_n = \sum_k \mathcal{B}_{n,k}$ . This is the Bell convention of the parent article, with  $\mathcal{Y}_n$  kept distinct from Bernoulli numbers.

**Theorem 8.1** (A master coefficient formula for finite  $q$ -products). *Let  $N_1, \dots, N_s \geq 0$ , and let  $a_\nu, \beta_\nu$  belong to the coefficient algebra. Define powers of unit series by formal logarithm and exponential. Put*

$$F(z) = \prod_{\nu=1}^s (a_\nu z; q)_{N_\nu}^{-\beta_\nu}, \quad L_j = \sum_{\nu=1}^s \beta_\nu a_\nu^j [N_\nu]_{q^j}. \quad (8.3)$$

Then

$$[z^n]F(z) = \frac{1}{n!} \mathcal{Y}_n(0!L_1, 1!L_2, \dots, (n-1)!L_n), \quad (8.4)$$

$$= \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum j m_j = n}} \prod_{j=1}^n \frac{L_j^{m_j}}{j^{m_j} m_j!}. \quad (8.5)$$

Every sum is finite and involves no auxiliary recurrence.

*Proof.* Expand the logarithm of each factor:

$$-\beta_\nu \sum_{r=0}^{N_\nu-1} \log(1 - a_\nu q^r z) = \sum_{j \geq 1} \frac{\beta_\nu a_\nu^j}{j} \left( \sum_{r=0}^{N_\nu-1} q^{rj} \right) z^j.$$

Summing gives  $\log F(z) = \sum_{j \geq 1} L_j z^j / j$ . Apply Equation (8.2), and then Equation (8.1). All rearrangements are coefficientwise finite because the outer  $q$ -products are finite.  $\square$

The list  $1, q, \dots, q^{N-1}$  is sometimes called a geometric alphabet. Its power sum is  $p_j = [N]_{q^j}$ . Logarithms convert multiplication of factors into addition of these power sums; Bell polynomials convert the result back to coefficients. No new “ $q$ -Bell polynomial” is necessary for this operation.

**Corollary 8.2** (Gaussian coefficients as ordinary Bell polynomials). *With  $p_j = [N]_{q^j}$ ,*

$$q^{\binom{k}{2}} \left[ \begin{matrix} N \\ k \end{matrix} \right]_q = \frac{1}{k!} \mathcal{Y}_k(p_1, -1!p_2, 2!p_3, \dots, (-1)^{k-1}(k-1)!p_k), \quad (8.6)$$

$$\left[ \begin{matrix} N+k-1 \\ k \end{matrix} \right]_q = \frac{1}{k!} \mathcal{Y}_k(p_1, 1!p_2, 2!p_3, \dots, (k-1)!p_k), \quad N \geq 1. \quad (8.7)$$

*Proof.* Apply Theorem 8.1 to  $(-z; q)_N$  and to  $1/(z; q)_N$ , then use Theorem 3.1.  $\square$

For instance,

$$q \left[ \begin{matrix} N \\ 2 \end{matrix} \right]_q = \frac{p_1^2 - p_2}{2}, \quad q^3 \left[ \begin{matrix} N \\ 3 \end{matrix} \right]_q = \frac{p_1^3 - 3p_1 p_2 + 2p_3}{6}. \quad (8.8)$$

These identities compress subset sums into power sums. The visible subtractions are compatible with the nonnegative polynomial model; they are a change of coordinates, not a loss of positivity.

## 8.2 Products, reciprocal products, and mixed parameters

For a formal parameter  $\lambda$ , Theorem 8.1 gives

$$[z^n](z; q)_N^{-\lambda} = \sum_{\sum j m_j = n} \prod_{j=1}^n \frac{\lambda^{m_j} [N]_{q^j}^{m_j}}{j^{m_j} m_j!}. \quad (8.9)$$

It is polynomial in  $\lambda$  of degree at most  $n$ . With  $\lambda$  a positive integer, one may instead expand each factor ordinarily:

$$[z^n](z; q)_N^{-\lambda} = \sum_{d_0 + \dots + d_{N-1} = n} q^{\sum_{j=0}^{N-1} j d_j} \prod_{j=0}^{N-1} \binom{\lambda + d_j - 1}{d_j}. \quad (8.10)$$

The proof is multiplication of  $(1 - q^j z)^{-\lambda}$ ; hence Equations (8.9) and (8.10) form an explicit identity between a partition sum and an ordinary-binomial weighted sum. More general mixed ratios require only the signed power sums  $L_j$  of Equation (8.3). In particular, negative  $\beta_\nu$  introduce numerator factors without changing the coefficient mechanism.

## 9 Jackson differentiation and geometric Taylor expansion

### 9.1 Coefficient extraction and Newton polynomials

Recall that

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x}, \quad D_q x^n = [n]_q x^{n-1}.$$

For a formal power series the expression at  $x = 0$  means its regular power-series extension. Consequently,

$$(D_q^n f)(0) = [n]_q! [x^n] f(x). \quad (9.1)$$

In this section  $q$  is generic; in analytic statements it is enough to exclude the roots of unity that make the displayed denominators vanish. The elementary operator formulae below agree with the conventions of [7, §17.2(iv)].

Set

$$P_k(x; a) = \prod_{j=0}^{k-1} (x - aq^j), \quad P_0(x; a) = 1.$$

Factoring the  $k - 1$  common factors in  $P_k(x; a) - P_k(qx; a)$  gives

$$D_q P_k(x; a) = [k]_q P_{k-1}(x; a). \quad (9.2)$$

Indeed  $P_k(qx; a) = q^{k-1}(qx - a)P_{k-1}(x; a)$ , whereas  $P_k(x; a) = (x - aq^{k-1})P_{k-1}(x; a)$ ; their difference is  $(1 - q^k)xP_{k-1}(x; a)$ .

**Theorem 9.1** (Polynomial geometric Taylor formula). *For every polynomial  $p$  of degree at most  $d$ ,*

$$p(x) = \sum_{k=0}^d \frac{(D_q^k p)(a)}{[k]_q!} P_k(x; a). \quad (9.3)$$

When  $a, qa, \dots, q^k a$  are distinct,

$$\frac{(D_q^k p)(a)}{[k]_q!} = p[a, qa, \dots, q^k a]. \quad (9.4)$$

The same equality holds for arbitrary functions defined on these nodes, with the Jackson derivative evaluated by its finite-difference formula.

*Proof.* The monic polynomials  $P_k$  are a basis. Apply  $D_q^r$  to an expansion  $p = \sum c_k P_k$  and put  $x = a$ . Terms with  $k < r$  vanish after differentiation, and terms with  $k > r$  contain the factor  $P_{k-r}(a; a) = 0$ . The remaining term is  $c_r [r]_q!$ . This proves Equation (9.3). Uniqueness of Newton interpolation proves Equation (9.4). For arbitrary node data choose their interpolating polynomial: both sides depend on exactly those data, not on the choice of an extension.  $\square$

No convergence assertion for an infinite geometric Taylor series has been made. Polynomial identities and finite interpolation require no such assertion.

### 9.2 An explicit finite-difference formula

Let  $T_q f(x) = f(qx)$ . Since  $T_q x^{-1} = q^{-1} x^{-1} T_q$ , induction gives

$$D_q^n = \frac{x^{-n}}{(1-q)^n} \prod_{r=0}^{n-1} (1 - q^{-r} T_q).$$



Expand this product by the finite  $q$ -binomial theorem, using reciprocity of Gaussian coefficients, to obtain

$$(D_q^n f)(x) = \frac{1}{(1-q)^n x^n} \sum_{j=0}^n (-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q f(q^j x). \quad (9.5)$$

Here  $x \neq 0$ ; for a regular function at zero the limiting or formal coefficient interpretation is used. The factor  $q^{-(n-1)j}$  is essential. Omitting it is a common consequence of treating multiplication by  $x^{-1}$  as though it commuted with  $T_q$ .

### 9.3 Leibniz and ordered multinomial formulae

**Theorem 9.2** (Jackson product rule). *For functions on the required geometric nodes,*

$$D_q^n (fg)(x) = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q (D_q^{n-j} f)(q^j x) (D_q^j g)(x). \quad (9.6)$$

For  $r$  factors its iterated form is

$$D_q^n \left( \prod_{i=1}^r f_i \right) (x) = \sum_{n_1 + \dots + n_r = n} \frac{[n]_q!}{\prod_i [n_i]_q!} \prod_{i=1}^r (D_q^{n_i} f_i) (q^{n_{i+1} + \dots + n_r} x). \quad (9.7)$$

*Proof.* The first-order identity is  $D_q(fg) = (D_q f)g + f(qx)D_q g$ . Also  $D_q(h(q^j x)) = q^j (D_q h)(q^j x)$ . Differentiate the right side of Equation (9.6). The coefficients of  $(D_q^{n+1-j} f)(q^j x)(D_q^j g)(x)$  add to  $q^j \begin{bmatrix} n \\ j \end{bmatrix}_q + \begin{bmatrix} n \\ j-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ j \end{bmatrix}_q$ . This proves the induction. Repeated application to the first factor and the remaining product yields Equation (9.7); the successive Gaussian coefficients telescope to the displayed multinomial.  $\square$

The same coefficient occurs in a genuinely noncommutative binomial formula. If two elements of an associative algebra satisfy  $YX = qXY$ , then

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^k Y^{n-k}. \quad (9.8)$$

Multiplication on the left gives the coefficient  $\begin{bmatrix} n \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n \\ k \end{bmatrix}_q$ , which is  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ ; this proves the formula. Its ordering convention is part of its statement. In contrast, for ordinary commuting  $x, y$ , the coefficients of  $(x + y)^n$  are ordinary binomial coefficients. Likewise  $e_q(x + y) = e_q(x)e_q(y)$  does not hold for commuting arguments in general.

## 10 An exact Jackson chain rule through divided differences

The preceding distinctions become especially important for composition. There is no general identity  $D_q(f \circ g) = (D_q f) \circ g \cdot D_q g$ . For  $f(u) = u^2$  and  $g(x) = x^2$ , its two sides would be  $[4]_q x^3$  and  $[2]_q^2 x^3$ , which differ for generic  $q$ . A correct finite chain rule uses divided differences. The general construction belongs to the divided-difference chain-rule literature [5]; the following proof also explains its relation to coefficient calculus.

**Theorem 10.1** (Path form of the divided-difference chain rule). *Let  $x_0, \dots, x_n$  be distinct and put  $y_i = g(x_i)$ . Suppose first that the  $y_i$  are distinct. For  $n \geq 1$ ,*

$$(f \circ g)[x_0, \dots, x_n] = \sum_{k=1}^n \sum_{0=i_0 < i_1 < \dots < i_k=n} f[y_{i_0}, \dots, y_{i_k}] \prod_{r=1}^k g[x_{i_{r-1}}, \dots, x_{i_r}]. \quad (10.1)$$

Whenever confluent limits exist, the same formula extends to coinciding  $y_i$  by replacing divided differences by their confluent values. For  $x_i = aq^i$  and  $a \neq 0$ , multiplication of the right side by  $[n]_q!$  gives  $(D_q^n(f \circ g))(a)$ .

*Proof.* We give the finite matrix argument in detail. Let  $T$  be the  $(n+1) \times (n+1)$  upper bidiagonal matrix whose diagonal is  $x_0, \dots, x_n$  and whose first superdiagonal consists of ones. Direct back substitution in  $(zI - T)R = I$  shows that

$$R_{ij} = \frac{1}{\prod_{h=i}^j (z - x_h)} \quad (i \leq j).$$

For a polynomial  $h$ , applying polynomial functional calculus, or expanding this rational expression by partial fractions, therefore gives

$$h(T)_{ij} = h[x_i, \dots, x_j]. \quad (10.2)$$

For completeness, the partial-fraction coefficient of  $1/(z - x_r)$  is  $\prod_{i \leq h \leq j, h \neq r} (x_r - x_h)^{-1}$ ; multiplying by  $h(x_r)$  and summing is exactly the explicit divided-difference formula.

More generally, let  $A$  be upper triangular with diagonal  $a_i$  and strict upper part  $N$ . The finite expansion

$$(zI - A)^{-1} = \sum_{k=0}^n ((zI - \text{diag } A)^{-1} N)^k (zI - \text{diag } A)^{-1}$$

shows that its  $(0, n)$  entry is a sum over increasing paths  $0 = i_0 < \dots < i_k = n$ . The contribution of a path is

$$\frac{A_{i_0 i_1} \cdots A_{i_{k-1} i_k}}{(z - a_{i_0}) \cdots (z - a_{i_k})}.$$

The same partial-fraction calculation gives

$$f(A)_{0n} = \sum_{\text{extpaths}} f[a_{i_0}, \dots, a_{i_k}] A_{i_0 i_1} \cdots A_{i_{k-1} i_k}.$$

Take  $A = g(T)$ , use Equation (10.2), and use  $f(g(T)) = (f \circ g)(T)$ . This proves Equation (10.1) for polynomials. For arbitrary data with distinct  $y_i$ , interpolate  $g$  on the  $x_i$  and  $f$  on the  $y_i$ ; every term is unchanged. Passage to confluent limits proves the stated extension when the necessary smoothness or analyticity is available. Finally apply Equation (9.4) to the geometric nodes.  $\square$

For  $n = 2$  the two paths give the particularly transparent identity

$$\begin{aligned} (f \circ g)[x_0, x_1, x_2] &= f[y_0, y_2]g[x_0, x_1, x_2] \\ &\quad + f[y_0, y_1, y_2]g[x_0, x_1]g[x_1, x_2]. \end{aligned} \quad (10.3)$$

As  $q \rightarrow 1$ , geometric nodes coalesce and divided differences become ordinary derivatives divided by factorials. Thus this is a finite-node precursor of ordinary Faà di Bruno, rather than an unjustified replacement of all factorials by  $q$ -factorials.

## 11 Composition in a $q$ -divided-power normalization

### 11.1 Products, powers, and logarithms

Write  $Q_n = [n]_q!$ , with  $Q_0 = 1$ , and represent an ordinary commuting formal series by

$$A(z) = \sum_{n \geq 0} a_n \frac{z^n}{Q_n}.$$

This is a change of coefficient normalization over  $\mathbb{Q}(q)$ , not a new notion of multiplication or substitution. In particular,

$$Q_n[z^n]A(z)B(z) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k b_{n-k}. \quad (11.1)$$

For several factors the coefficients are Gaussian multinomials. This follows immediately by multiplying monomials and using  $Q_n/(Q_k Q_{n-k}) = \begin{bmatrix} n \\ k \end{bmatrix}_q$ .

If  $a_0 = 1$ , the ordinary generalized binomial theorem gives the following closed profile formula. In every profile below  $m_j \geq 0$ ,  $\sum_{j=1}^n j m_j = n$ , and  $M = \sum_j m_j$ :

$$Q_n[z^n]A(z)^\lambda = Q_n \sum_{\sum j m_j = n} \frac{\lambda^{\underline{M}}}{\prod_j m_j!} \prod_{j=1}^n \left( \frac{a_j}{Q_j} \right)^{m_j}. \quad (11.2)$$

Here  $\lambda^{\underline{M}} = \lambda(\lambda-1)\cdots(\lambda-M+1)$  is an *ordinary* falling factorial. For  $n \geq 1$  the corresponding logarithm formula is

$$Q_n[z^n] \log A(z) = Q_n \sum_{\sum j m_j = n} \frac{(-1)^{M-1} (M-1)!}{\prod_j m_j!} \prod_j \left( \frac{a_j}{Q_j} \right)^{m_j}. \quad (11.3)$$

To prove both identities, put  $U = A - 1$  and expand  $(1+U)^\lambda = \sum_{M \geq 0} \lambda^{\underline{M}} U^M / M!$  or  $\log(1+U) = \sum_{M \geq 1} (-1)^{M-1} U^M / M$ . The multinomial theorem for  $U^M$  supplies  $M! / \prod m_j!$ . Since  $U$  has positive order, only finitely many terms contribute to each coefficient.

## 11.2 A precise partial Bell normalization

**Definition 11.1.** For  $G(z) = \sum_{j \geq 1} g_j z^j / Q_j$ , define

$$\mathcal{C}_{n,k}^{(q)}(g_1, g_2, \dots) = \frac{Q_n}{Q_k} [z^n] G(z)^k. \quad (11.4)$$

Set  $\mathcal{C}_{0,0}^{(q)} = 1$  and  $\mathcal{C}_{n,0}^{(q)} = 0$  for  $n > 0$ .

This notation fixes one particular  $q$ -normalization; it does not identify it with every family called a “ $q$ -Bell polynomial” in the literature. A direct multinomial expansion yields

$$\mathcal{C}_{n,k}^{(q)}(g) = \frac{Q_n}{Q_k} \sum_{\substack{\sum j m_j = n \\ \sum m_j = k}} \frac{k!}{\prod_j m_j!} \prod_j \left( \frac{g_j}{Q_j} \right)^{m_j} \quad (11.5)$$

$$= \frac{Q_n}{Q_k} \frac{k!}{n!} \mathcal{B}_{n,k} \left( \frac{1!g_1}{Q_1}, \frac{2!g_2}{Q_2}, \dots \right). \quad (11.6)$$

If  $F(u) = \sum_{k \geq 0} f_k u^k / Q_k$ , then

$$Q_n[z^n]F(G(z)) = \sum_{k=0}^n f_k \mathcal{C}_{n,k}^{(q)}(g). \quad (11.7)$$

Equations (11.5)–(11.7) are proved simply by substituting and collecting coefficients. Together with Equation (9.1) they give an at-zero Jackson derivative formula for formal composition. They do *not* give a shifted Jackson chain rule at a nonzero point.

There is an instructive warning already at degree three:

$$\mathcal{C}_{3,2}^{(q)}(g) = \frac{2[3]_q}{[2]_q} g_1 g_2. \quad (11.8)$$

This coefficient is rational, not polynomial, in  $q$ . Its  $q \rightarrow 1$  limit is  $3g_1 g_2$ , the ordinary partial Bell polynomial. A combinatorial family with polynomial  $q$ -weights requires additional structure; it does not follow from factorial renormalization alone.

### 11.3 Substitution matrices and a normalized Riordan group

For a series  $F$  of positive order, let  $C(F)$  be the lower triangular matrix with entries

$$C(F)_{n,k} = \frac{Q_n}{Q_k} [z^n] F(z)^k.$$

Substitution in two stages proves

$$C(F \circ G) = C(G)C(F). \quad (11.9)$$

The order is important: the inner series occurs in the left matrix. All matrix entries are finite sums, so the equation is meaningful for formal series without analytic convergence assumptions.

More generally, set

$$R(g, f)_{n,k} = \frac{Q_n}{Q_k} [z^n] g(z) f(z)^k, \quad f(0) = 0. \quad (11.10)$$

When  $g(0)$  and  $f'(0)$  are units, these matrices form a group and satisfy

$$R(g, f)R(h, \ell) = R(g(h \circ f), \ell \circ f), \quad (11.11)$$

$$R(g, f)^{-1} = R\left(\frac{1}{g \circ \bar{f}}, \bar{f}\right), \quad (11.12)$$

where  $\bar{f} = f^{-1}$  is the ordinary compositional inverse. To prove the product law, expand  $h(u)\ell(u)^k$  in powers of  $u$ , substitute  $u = f(z)$ , and observe that the intermediate  $Q_j$  factors cancel. The inverse then follows by substitution into the product law. Equivalently, this is diagonal conjugation of the ordinary power-series Riordan representation by  $\text{diag}(Q_n)$ . Thus it extends the parent's matrix viewpoint without introducing a new associative law.

As a useful check,

$$R(e_q(z), z)_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

and its inverse is  $R(E_q(-z), z)$ . This recovers Gaussian inversion, including the factor  $(-1)^{n-k} q^{\binom{n-k}{2}}$ . If polynomials  $P_n(x)$  are defined by  $g(z)e_q(xz) = \sum P_n(x)z^n/Q_n$ , then  $D_q P_n(x) = [n]_q P_{n-1}(x)$ : apply  $D_q$  in  $x$  to the generating function and compare coefficients. This supplies the corresponding  $q$ -Appell lowering property with no ambiguity about normalization.

## 12 Ordinary Lagrange inversion with Gaussian input

### 12.1 A closed coefficient theorem

Let  $K$  be a characteristic-zero field containing the needed parameters, and let  $\Phi(w) \in 1 + wK[[w]]$ . There is a unique  $g(z) \in zK[[z]]$  satisfying  $g = z\Phi(g)$ : its coefficient of degree  $n$  is fixed by the previous coefficients. The following formula eliminates that recurrence.

**Theorem 12.1** (Lagrange–Bell coefficient formula). *For  $1 \leq r \leq n$ ,*

$$[z^n]g(z)^r = \frac{r}{n} [w^{n-r}] \Phi(w)^n. \quad (12.1)$$

*In particular, if*

$$\Phi(w) = \prod_{\nu=1}^s (a_\nu w; q)_{N_\nu}^{-\beta_\nu}, \quad L_j = \sum_{\nu=1}^s \beta_\nu a_\nu^j [N_\nu]_{q^j}, \quad (12.2)$$

then

$$[z^n]g(z)^r = \frac{r}{n(n-r)!} \mathcal{Y}_{n-r}(nL_1, n1!L_2, n2!L_3, \dots) \quad (12.3)$$

$$= \frac{r}{n} \sum_{\sum j m_j = n-r} \prod_{j=1}^{n-r} \frac{(nL_j/j)^{m_j}}{m_j!}. \quad (12.4)$$

An empty profile contributes one when  $n = r$ .

*Proof.* Put  $f(w) = w/\Phi(w)$ ; it is the compositional inverse of  $g$ . Formal residue substitution gives

$$\begin{aligned} [z^n]g^r &= \text{Res}_w w^r f(w)^{-n-1} f'(w) dw \\ &= -\frac{1}{n} \text{Res}_w w^r \frac{d}{dw} f(w)^{-n} dw = \frac{r}{n} \text{Res}_w w^{r-1} f(w)^{-n} dw. \end{aligned}$$

The last equality uses that the residue of a formal derivative is zero. This is Equation (12.1). Residue substitution itself follows by linearity from monomials: for exponents different from  $-1$  the substituted expression is a derivative, and  $f'/f$  has residue one. Finally, Equation (12.2) gives  $\log \Phi = \sum_{j \geq 1} L_j w^j / j$ . Apply the proved Bell expansion of an exponential to  $\Phi^n$ . This proves both closed forms.  $\square$

The prefactor is  $r/n$ , not  $[r]_q/[n]_q$ . Ordinary composition and its formal residue Jacobian have not changed. Replacing this prefactor by a  $q$ -ratio generally produces an incorrect inverse.

**Example 12.2** (A cubic inverse). The inverse at zero of  $f(w) = w(1-w)(1-qw)$  begins

$$\begin{aligned} g(z) &= z + (1+q)z^2 + (2+3q+2q^2)z^3 \\ &\quad + (5+10q+10q^2+5q^3)z^4 \\ &\quad + (14+35q+45q^2+35q^3+14q^4)z^5 + O(z^6). \end{aligned}$$

Every coefficient is obtained from Equation (12.4) using  $L_j = 1 + q^j$ . Equivalently,  $[z^n]g = n^{-1} \sum_{a+b=n-1} \binom{n+a-1}{a} \binom{n+b-1}{b} q^b$ . This is a finite binomial sum, and substitution into the cubic supplies a separate check.

## 12.2 Optional-slot trees and a Gaussian divisibility identity

A second choice of input reveals polynomial positivity that is not obvious from Lagrange's factor  $1/n$ .

**Theorem 12.3** (Gaussian slot-tree coefficients). *Fix an integer  $N \geq 1$  and define  $T_N(z)$  by*

$$T_N(z) = z \prod_{j=0}^{N-1} (1 + q^j T_N(z)). \quad (12.5)$$

For  $n \geq 1$ , its coefficient  $C_{N,n}(q) = [z^n]T_N(z)$  satisfies

$$C_{N,n}(q) = \frac{1}{n} \sum_{\substack{k_1 + \dots + k_n = n-1 \\ 0 \leq k_i \leq N}} q^{\sum_{i=1}^n \binom{k_i}{2}} \prod_{i=1}^n \begin{bmatrix} N \\ k_i \end{bmatrix}_q. \quad (12.6)$$

Moreover,

$$C_{N,n}(q) \in \mathbb{Z}_{\geq 0}[q], \quad (12.7)$$

$$q^{(N-1)(n-1)} C_{N,n}(q^{-1}) = C_{N,n}(q), \quad (12.8)$$

$$C_{N,n}(1) = \frac{1}{n} \binom{Nn}{n-1} = \frac{1}{(N-1)n+1} \binom{Nn}{n}. \quad (12.9)$$

Thus the numerator of Equation (12.6) is divisible by  $n$  coefficientwise in  $\mathbb{Z}[q]$ .

*Proof.* Use Lagrange inversion with  $\Phi(w) = (-w; q)_N$ , expand each of the  $n$  factors by the finite  $q$ -binomial theorem, and extract degree  $n - 1$ . This proves Equation (12.6). For integrality and positivity, consider rooted trees in which each vertex has  $N$  labeled child slots  $0, \dots, N - 1$ , each either empty or occupied by one subtree. Give an edge in slot  $j$  weight  $q^j$ , and give each vertex weight  $z$ . Decomposing at the root gives Equation (12.5). There are finitely many such trees of each size; their weight sum is a polynomial with nonnegative integer coefficients. Uniqueness of the formal solution identifies it with  $T_N$ .

Reverse every slot label  $j \mapsto N - 1 - j$ . A tree of  $n$  vertices has  $n - 1$  edges, so its exponent  $w$  changes to  $(N - 1)(n - 1) - w$ . This weight-reversing bijection proves reciprocity. Finally put  $q = 1$  in Lagrange's formula:  $\Phi(w)^n = (1 + w)^{Nn}$ . The equality of the two ordinary binomial expressions is a direct factorial cancellation.  $\square$

For  $N = 2$  an especially simple form is

$$C_{2,n}(q) = \frac{1}{n} \sum_{j=0}^{n-1} \binom{n}{j} \binom{n}{j+1} q^j. \quad (12.10)$$

For example  $C_{2,3} = 1 + 3q + q^2$ . The formula follows by extracting  $w^{n-1}$  from  $(1 + w)^n(1 + qw)^n$ . These are Narayana polynomials under the displayed normalization. They are not the same  $q$ -Catalan family as a Gaussian quotient such as  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q / [n + 1]_q$ . The name “ $q$ -Catalan” by itself does not specify a weight statistic or a unique deformation.

## 13 Parameter derivatives and weighted Gaussian sums

### 13.1 Differentiating a finite Pochhammer product

Let  $F(a) = (a; q)_N$ . Away from its zeros, direct logarithmic differentiation gives, for  $r \geq 1$ ,

$$L_r(a) := \frac{d^r}{da^r} \log F(a) = -(r - 1)! \sum_{j=0}^{N-1} \frac{q^{jr}}{(1 - aq^j)^r}. \quad (13.1)$$

Since  $F(a + h)/F(a) = \exp(\sum_{r \geq 1} L_r(a) h^r / r!)$ , the Bell coefficient identity immediately gives

$$\frac{d^m}{da^m} (a; q)_N = (a; q)_N \mathcal{Y}_m(L_1(a), \dots, L_m(a)). \quad (13.2)$$

The apparent poles cancel on the left; the displayed quotient form is used only where its factors are defined. For  $(a; q)_N^{-\lambda}$ , replace every  $L_r$  by  $-\lambda L_r$ . In this way all derivative orders are finite profile sums, without repeatedly differentiating a product.

The Euler operator  $a\partial_a$  brings the ordinary Stirling numbers back into the picture:

$$\begin{aligned} \sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q k^s a^k &= (a\partial_a)^s (a; q)_N \\ &= \sum_{r=0}^s \left\{ \begin{matrix} s \\ r \end{matrix} \right\} a^r \partial_a^r (a; q)_N. \end{aligned} \quad (13.3)$$

For  $s = 0$  use  $0^0 = 1$ . The first equality follows by applying the operator to the finite  $q$ -binomial theorem. The second follows from  $(a\partial_a)^s = \sum_r \left\{ \begin{matrix} s \\ r \end{matrix} \right\} a^r \partial_a^r$ , proved either on each monomial or by the ordinary Stirling recurrence. Combining Equations (13.2) and (13.3) gives closed formulae for all polynomially weighted signed Gaussian sums. The Stirling numbers here are ordinary because  $a\partial_a$ , rather than a Jackson operator, is being used.

### 13.2 Euler derivatives in the base $q$

Write  $\theta = q\partial_q$  and define the rational functions

$$R_r(x) = \left(x \frac{d}{dx}\right)^{r-1} \frac{x}{1-x} = \text{Li}_{1-r}(x), \quad r \geq 1. \quad (13.4)$$

The polylogarithm notation is optional: the following finite expression makes the rational nature explicit:

$$R_r(x) = \sum_{\ell=1}^r (\ell-1)! \left\{ \begin{matrix} r \\ \ell \end{matrix} \right\} \frac{x^\ell}{(1-x)^\ell}. \quad (13.5)$$

To verify it, note that for  $|x| < 1$  the left side is  $\sum_{m \geq 1} m^{r-1} x^m$ . Alternatively, differentiating the right side and using the Stirling recurrence proves the same formula from  $R_1 = x/(1-x)$ ; rational continuation removes the restriction  $|x| < 1$ . For  $r \geq 2$  the Eulerian form is

$$R_r(x) = \frac{x A_{r-1}(x, 1)}{(1-x)^r}, \quad (13.6)$$

by the ordinary specialization of the proved Eulerian generating function. Thus both Stirling and Eulerian bases organize derivatives of the same  $q$ -product.

For  $0 \leq k \leq n$  define, away from zeros and poles of the factors,

$$K_r(n, k; q) = \sum_{i=1}^k \left( i^r R_r(q^i) - (n-k+i)^r R_r(q^{n-k+i}) \right). \quad (13.7)$$

Logarithmic differentiation of the Gaussian product gives  $K_r = \theta^r \log \begin{bmatrix} n \\ k \end{bmatrix}_q$ , hence

$$\theta^m \begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q \mathcal{Y}_m(K_1, \dots, K_m). \quad (13.8)$$

Indeed  $q \mapsto qe^t$  converts  $\theta$  into  $\partial_t$ , reducing this to the ordinary Bell exponential formula. The signed ordinary first-kind Stirling numbers  $s(m, r)$  convert Euler derivatives back to ordinary ones:

$$\partial_q^m \begin{bmatrix} n \\ k \end{bmatrix}_q = q^{-m} \sum_{r=0}^m s(m, r) \theta^r \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (13.9)$$

This follows from  $q^m \partial_q^m = \theta(\theta-1) \cdots (\theta-m+1)$ . At  $q=1$  the separate terms in Equation (13.7) are singular, even though the Gaussian polynomial is not. A different local representation gives exact finite values without cancellation of infinities.

## 14 All Taylor coefficients at $q=1$

### 14.1 Bernoulli cumulants

Use Bernoulli numbers  $B_j$  with convention

$$\frac{t}{e^t - 1} = \sum_{j \geq 0} B_j \frac{t^j}{j!}, \quad B_1 = -\frac{1}{2}. \quad (14.1)$$

This is the standard convention in [10, §24.2]. Put

$$d = k(n-k), \quad \Delta_s(n, k) = \sum_{j=1}^n j^s - \sum_{j=1}^k j^s - \sum_{j=1}^{n-k} j^s. \quad (14.2)$$

Empty sums are zero. We state the result for  $n \geq 1$ ; the case  $n=0$  is the constant polynomial one.

**Theorem 14.1** (The exponential-coordinate expansion). *For  $0 \leq k \leq n$  and  $q = e^t$ ,*

$$\log \frac{[n]_{e^t}}{\binom{n}{k}} = \frac{d}{2}t + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}(n, k)}{2r(2r)!} t^{2r}. \quad (14.3)$$

*The logarithm is its analytic germ at  $t = 0$ . The series converges for  $|t| < 2\pi/n$  and is also a formal identity. Define*

$$\kappa_1 = \frac{d}{2}, \quad \kappa_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k), \quad \kappa_{2r+1} = 0 \quad (r \geq 1). \quad (14.4)$$

*Then*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^t} = \binom{n}{k} \sum_{m \geq 0} \mathcal{Y}_m(\kappa_1, \dots, \kappa_m) \frac{t^m}{m!}. \quad (14.5)$$

*In particular, every ordinary derivative at  $q = 1$  is given by the finite sum*

$$\left. \partial_q^m \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \right|_{q=1} = \binom{n}{k} \sum_{j=0}^m s(m, j) \mathcal{Y}_j(\kappa_1, \dots, \kappa_j). \quad (14.6)$$

*Proof.* Differentiating and using Equation (14.1), then integrating with zero constant term, gives

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} u^{2r}.$$

The nearest nonzero zeros of  $e^u - 1$  are  $\pm 2\pi i$ , so this local logarithm is analytic for  $|u| < 2\pi$ . In the Gaussian product cancel the factors of  $t$  and the integer factorials, and add these logarithms with  $u = jt$ , with signs prescribed by numerator and denominators. The linear term is  $\Delta_1 t/2 = dt/2$ , giving Equation (14.3). Exponentiation gives Equation (14.5). Finally Equation (13.9) evaluated at one gives Equation (14.6). Every step is valid as a formal series, and the preceding disk also gives the asserted analytic validity.  $\square$

All quantities in these formulae are finite. For example, if  $B_m(x)$  denotes the Bernoulli polynomial defined by  $te^{xt}/(e^t - 1) = \sum B_m(x)t^m/m!$ , then

$$\Delta_s(n, k) = \frac{B_{s+1}(n+1) - B_{s+1}(k+1) - B_{s+1}(n-k+1) + B_{s+1}(1)}{s+1}. \quad (14.7)$$

To prove this power-sum formula, subtract the generating functions at  $x+1$  and  $x$  to get  $B_{s+1}(x+1) - B_{s+1}(x) = (s+1)x^s$ , then telescope.

## 14.2 Area statistics and exact central moments

Choose uniformly a  $k$ -subset of  $\{0, 1, \dots, n-1\}$  and let  $X$  be the sum of its elements minus  $\binom{k}{2}$ . The subset formula shows that  $\mathbb{E}e^{tX} = [n]_{e^t} / \binom{n}{k}$ . Thus Equation (14.4) really is a cumulant formula for this finite distribution. In particular,

$$\mathbb{E}X = \frac{d}{2}, \quad \text{Var}(X) = \frac{d(n+1)}{12}, \quad \mathbb{E}(X - d/2)^4 = 3 \left( \frac{d(n+1)}{12} \right)^2 - \frac{\Delta_4}{120}. \quad (14.8)$$

Here  $\Delta_2 = d(n+1)$ ,  $B_2 = 1/6$ , and  $B_4 = -1/30$ . Every odd centered moment is zero, since the complement-in-the-rectangle bijection sends  $X$  to  $d - X$ ; equivalently the centered logarithm in Equation (14.3) is even. Higher centered moments are obtained by setting the first Bell argument to zero.



The first two ordinary derivatives illustrate why Euler derivatives and ordinary derivatives must not be confused:

$$\partial_q \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \Big|_1 = \binom{n}{k} \frac{d}{2}, \quad (14.9)$$

$$\partial_q^2 \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \Big|_1 = \binom{n}{k} \left( \frac{d^2}{4} + \frac{d(n+1)}{12} - \frac{d}{2} \right). \quad (14.10)$$

The subtraction of  $d/2$  converts the second raw moment into the second falling-factorial moment. These exact identities require no limiting probability theorem.

## 15 Roots of unity: polynomial evaluation and local jets

### 15.1 Cyclotomic factorization

Let  $\Phi_\ell(q)$  be the  $\ell$ th cyclotomic polynomial, and define

$$\nu_\ell(n, k) = \left\lfloor \frac{n}{\ell} \right\rfloor - \left\lfloor \frac{k}{\ell} \right\rfloor - \left\lfloor \frac{n-k}{\ell} \right\rfloor. \quad (15.1)$$

Factoring each  $1 - q^j$  into cyclotomic factors gives

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \prod_{\ell=2}^n \Phi_\ell(q)^{\nu_\ell(n, k)}, \quad \nu_\ell(n, k) \in \{0, 1\}. \quad (15.2)$$

Indeed the multiplicity in  $(q; q)_m$  is  $\lfloor m/\ell \rfloor$ , and  $\lfloor u+v \rfloor - \lfloor u \rfloor - \lfloor v \rfloor$  is zero or one. The  $\ell = 1$  exponent is zero. This proves not only regularity at roots of unity but also that every complex zero of a nonconstant Gaussian polynomial is simple. The multiplicity of a *particular primitive root* is one or zero; several different cyclotomic factors can of course occur.

### 15.2 The $q$ -Lucas evaluation

**Theorem 15.1** (Gaussian Lucas formula). *Let  $\zeta$  be a primitive  $\ell$ th root of unity,  $\ell \geq 2$ , and write  $n = a\ell + b$ ,  $k = c\ell + d$ , where  $0 \leq b, d < \ell$ . For  $0 \leq k \leq n$ ,*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_\zeta = \binom{a}{c} \left[ \begin{matrix} b \\ d \end{matrix} \right]_\zeta. \quad (15.3)$$

As usual the right side is zero if  $d > b$ . Values are taken in the Gaussian polynomial, not in an uncanceled quotient  $0/0$ .

*Proof.* At a primitive root,

$$\prod_{j=0}^{\ell-1} (1 + \zeta^j z) = 1 - (-z)^\ell.$$

Split the  $n$  factors into  $a$  full blocks and a block of length  $b$ :

$$\prod_{j=0}^{n-1} (1 + \zeta^j z) = (1 - (-z)^\ell)^a \prod_{j=0}^{b-1} (1 + \zeta^j z).$$

The coefficient of  $z^{c\ell+d}$  on the right is zero if  $d > b$ ; otherwise it is  $\binom{a}{c} (-1)^{c(\ell+1)} \zeta^{\binom{d}{2}} \left[ \begin{matrix} b \\ d \end{matrix} \right]_\zeta$ . The coefficient on the left is  $\zeta^{\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_\zeta$ . To cancel the phase, observe

$$\zeta^{\binom{k}{2} - \binom{d}{2}} = (-1)^{c(\ell+1)}.$$

For odd  $\ell$  the exponent difference is a multiple of  $\ell$ ; for even  $\ell$  use  $\zeta^{\ell/2} = -1$  and the fact that  $c\ell + 2d - 1$  is odd. This proves Equation (15.3) including the sign.  $\square$

This is a classical identity; see [6] for its place among Gaussian identities and extensions to negative arguments. At  $q = -1$  it becomes

$$\begin{bmatrix} n \\ k \end{bmatrix}_{-1} = \begin{cases} 0, & n \text{ even and } k \text{ odd,} \\ \begin{bmatrix} \lfloor n/2 \rfloor \\ \lfloor k/2 \rfloor \end{bmatrix}, & \text{otherwise.} \end{cases} \quad (15.4)$$

### 15.3 All local coefficients after removing a possible zero

Fix  $\zeta$  primitive of order  $\ell \geq 2$ , let  $\nu = \nu_\ell(n, k)$ , and put

$$H(q) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\Phi_\ell(q)^\nu}.$$

When  $\ell > n$ , take  $\nu = 0$ . Then  $H(\zeta) \neq 0$ . The following is a finite algebraic prescription for the full Taylor expansion at  $\zeta$ :

$$L_r = (-1)^{r-1} (r-1)! \sum_{\substack{2 \leq d \leq n \\ d \neq \ell}} \nu_d(n, k) \sum_{\substack{\omega \text{ primitive} \\ d\text{th root}}} (\zeta - \omega)^{-r}, \quad (15.5)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_{\zeta+h} = \Phi_\ell(\zeta + h)^\nu H(\zeta) \sum_{m \geq 0} \mathcal{Y}_m(L_1, \dots, L_m) \frac{h^m}{m!}. \quad (15.6)$$

To prove this, factor  $H$  into the linear factors  $q - \omega$  of Equation (15.2). Its  $r$ th logarithmic derivative at  $\zeta$  is exactly Equation (15.5). Exponentiate its Taylor series using Bell polynomials and multiply back the removed factor. The series is analytic up to the nearest zero of  $H$ ; it is also an identity of formal Taylor series over the cyclotomic field. Each prescribed coefficient is a finite sum, even though the full Taylor series for  $H$  written as an exponential is infinite.

If  $\nu = 1$ , the first nonzero coefficient is particularly simple:

$$\partial_q \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\zeta} = \Phi'_\ell(\zeta) H(\zeta). \quad (15.7)$$

Higher coefficients are finite convolutions of the polynomial  $\Phi_\ell(\zeta + h)$  with the Bell coefficients in Equation (15.6). Thus root evaluation, multiplicity, and all local derivatives fit the same coefficient calculus, provided that a zero is removed before taking its logarithm.

## 16 Three distinct asymptotic regimes

A formula that is uniform for fixed  $q$  need not be uniform as  $q \rightarrow 1$ . This section separates three useful regimes instead of silently mixing them. Finite coefficient formulae give an exact expansion in the first, infinite products give convergent exponential corrections in the second, and Euler–Maclaurin gives a uniform Poincaré expansion in the third. The last calculation extends the parent’s asymptotic coefficient machinery to a Gaussian input.

### 16.1 Fixed lower index and fixed base

For fixed  $k \geq 0$  and  $0 < |q| < 1$ , the finite  $q$ -binomial theorem gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k} \sum_{j=0}^k (-1)^j q^{j(n-k+1) + \binom{j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q, \quad n \geq k. \quad (16.1)$$

This is simply the product  $(q^{n-k+1}; q)_k / (q; q)_k$ . It is an *exact finite* expansion in powers of  $q^n$ , not an infinite asymptotic ansatz. In particular,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q \longrightarrow \frac{1}{(q; q)_k} \quad (n \rightarrow \infty).$$

For  $k \geq 1$  the first correction is  $-q^{n-k+1}[k]_q / (q; q)_k$ , with the remaining finite terms explicitly given by Equation (16.1).

## 16.2 Both sides large and fixed base

Fix  $0 < |q| < 1$ , and let  $k, n - k \rightarrow \infty$ . Put  $a = q^{k+1}$ ,  $b = q^{n-k+1}$ , and  $c = q^{n+1} = ab/q$ . Since  $(q; q)_m = (q; q)_\infty / (q^{m+1}; q)_\infty$ ,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \frac{(a; q)_\infty (b; q)_\infty}{(c; q)_\infty}. \quad (16.2)$$

The infinite  $q$ -binomial theorem therefore gives the exact convergent expansion

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \sum_{r,s,t \geq 0} \frac{(-1)^{r+s} q^{\binom{r}{2} + \binom{s}{2}} a^r b^s c^t}{(q; q)_r (q; q)_s (q; q)_t}. \quad (16.3)$$

The two numerator series converge absolutely for every finite  $a, b$ ; the reciprocal series converges absolutely for  $|c| < 1$ , which holds here. Hence their product and rearrangement are justified. The logarithm of the relative correction is

$$\log((q; q)_\infty \left[ \begin{matrix} n \\ k \end{matrix} \right]_q) = - \sum_{j \geq 1} \frac{a^j + b^j - c^j}{j(1 - q^j)}. \quad (16.4)$$

It follows by expanding the logarithm of each product; the absolute convergence for  $|a|, |b|, |c| < 1$  justifies interchanging sums. In particular,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_\infty} \left( 1 - \frac{a+b}{1-q} + O(|a|^2 + |b|^2) \right). \quad (16.5)$$

The constant may depend on the fixed base  $q$ ; the  $c$  term is of order  $ab$ . Higher corrections can be generated either by the triple sum or by the ordinary Bell expansion of Equation (16.4). None of these fixed-base formulae asserts uniformity near  $q = 1$ .

## 16.3 The double scaling $q = e^{-\tau/n}$

Let  $k = k_n$  be an integer, set  $\alpha = k/n$  and  $b = 1 - \alpha$ , and assume

$$\varepsilon \leq \alpha \leq 1 - \varepsilon, \quad 0 < \tau_0 \leq \tau \leq \tau_1 < \infty, \quad q = e^{-\tau/n}. \quad (16.6)$$

Thus  $\alpha$  denotes the actual ratio  $k/n$ , including any rounding; there is no unrecorded floor correction. Define a smooth function on  $[0, 1]$  by

$$h_\tau(x) = \log \frac{1 - e^{-\tau x}}{\tau x}, \quad h_\tau(0) = 0, \quad (16.7)$$

and set

$$\mathcal{S}_\tau(\alpha) = -\alpha \log \alpha - b \log b + \int_0^1 h_\tau(x) dx - \int_0^\alpha h_\tau(x) dx - \int_0^b h_\tau(x) dx \quad (16.8)$$

$$= \frac{\text{Li}_2(e^{-\tau}) - \text{Li}_2(e^{-\tau\alpha}) - \text{Li}_2(e^{-\tau b}) + \pi^2/6}{\tau}. \quad (16.9)$$

The equality follows from  $\frac{d}{dx} \text{Li}_2(e^{-\tau x}) = \tau \log(1 - e^{-\tau x})$  and elementary integration of  $\log(\tau x)$ . One can take Equation (16.8) as the definition without using a special function.

**Theorem 16.1** (Uniform all-order logarithmic expansion). *Under Equation (16.6), for every fixed  $M \geq 0$ ,*

$$\begin{aligned} \log \left[ \frac{n}{k} \right]_{e^{-\tau/n}} &= n \mathcal{S}_\tau(\alpha) - \frac{1}{2} \log(2\pi n \alpha b) + \frac{1}{2} (h_\tau(1) - h_\tau(\alpha) - h_\tau(b)) \\ &\quad + \sum_{r=1}^M \frac{C_{2r-1}(\alpha, \tau)}{n^{2r-1}} + O(n^{-(2M+1)}), \end{aligned} \quad (16.10)$$

with an error uniform on the indicated compact parameter set. The coefficients are explicit:

$$\begin{aligned} C_{2r-1}(\alpha, \tau) &= \frac{B_{2r}}{2r(2r-1)} (1 - \alpha^{1-2r} - b^{1-2r}) \\ &\quad + \frac{B_{2r}}{(2r)!} (h_\tau^{(2r-1)}(1) - h_\tau^{(2r-1)}(\alpha) - h_\tau^{(2r-1)}(b) + h_\tau^{(2r-1)}(0)). \end{aligned} \quad (16.11)$$

There are no even negative powers in this logarithmic expansion.

*Proof.* The product identity has the exact decomposition

$$\log \left[ \frac{n}{k} \right]_{e^{-\tau/n}} = \log \binom{n}{k} + \sum_{j=1}^n h_\tau(j/n) - \sum_{j=1}^k h_\tau(j/n) - \sum_{j=1}^{n-k} h_\tau(j/n). \quad (16.12)$$

Indeed  $1 - e^{-\tau j/n} = (\tau j/n) e^{h_\tau(j/n)}$ , and the elementary factors cancel to the ordinary binomial coefficient.

For an endpoint  $u \in \{1, \alpha, b\}$  with  $un$  integral, Euler–Maclaurin on mesh  $1/n$  reads

$$\begin{aligned} \sum_{j=1}^{un} h_\tau(j/n) &= n \int_0^u h_\tau(x) dx + \frac{h_\tau(u) - h_\tau(0)}{2} \\ &\quad + \sum_{r=1}^M \frac{B_{2r}}{(2r)! n^{2r-1}} (h_\tau^{(2r-1)}(u) - h_\tau^{(2r-1)}(0)) + O(n^{-(2M+1)}). \end{aligned}$$

To justify this error with the displayed truncation, apply the usual periodic-Bernoulli remainder formula one order farther, then absorb both the next boundary term and the remainder into  $O(n^{-(2M+1)})$ . The derivatives of  $h_\tau$  of every fixed order are uniformly bounded on  $[0, 1] \times [\tau_0, \tau_1]$ , because its apparent singularity at zero is removable. Hence the bounds are uniform. The underlying Euler–Maclaurin formula follows by integration by parts successively with periodic Bernoulli functions; its standard remainder form is recorded in [11, §2.10(i)].

Finally the factorial Stirling expansion [12, §5.11] gives

$$\begin{aligned} \log \binom{n}{k} &= n(-\alpha \log \alpha - b \log b) - \frac{1}{2} \log(2\pi n \alpha b) \\ &\quad + \sum_{r=1}^M \frac{B_{2r}}{2r(2r-1)n^{2r-1}} (1 - \alpha^{1-2r} - b^{1-2r}) + O(n^{-(2M+1)}). \end{aligned}$$

Its remainder is uniform since  $k, n - k \geq \varepsilon n$ . Insert these three Euler–Maclaurin expansions and the Stirling expansion into Equation (16.12); collect the integral, endpoint, and odd-power terms. This gives precisely Equations (16.10) and (16.11).  $\square$

## 16.4 Explicit derivatives and multiplicative coefficients

No recursive differentiation is needed in Equation (16.11). For  $x > 0$  and  $s \geq 1$ ,

$$h_\tau^{(s)}(x) = (-1)^{s-1} \left( \tau^s \text{Li}_{1-s}(e^{-\tau x}) - \frac{(s-1)!}{x^s} \right). \quad (16.13)$$

It follows by differentiating  $\log(1 - e^{-\tau x}) - \log(\tau x)$  and using Equation (13.4). At zero the nonsingular values are

$$h'_\tau(0) = -\frac{\tau}{2}, \quad h^{(2r)}_\tau(0) = \frac{B_{2r}\tau^{2r}}{2r}, \quad h^{(2r+1)}_\tau(0) = 0 \quad (r \geq 1), \quad (16.14)$$

by the Bernoulli logarithm expansion. Equations (13.5), (16.13) and (16.14) therefore make every  $C_j$  a finite expression in rational functions of  $e^{-\tau}$ ,  $e^{-\tau\alpha}$ ,  $e^{-\tau b}$  and the parameters.

Exponentiating the leading terms simplifies the prefactor:

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-\tau/n}} \sim e^{n\mathcal{S}_\tau(\alpha)} \sqrt{\frac{\tau(1 - e^{-\tau})}{2\pi n(1 - e^{-\tau\alpha})(1 - e^{-\tau b})}} \sum_{m \geq 0} \frac{H_m(\alpha, \tau)}{n^m}, \quad (16.15)$$

where  $C_{2r} = 0$  and

$$H_m = \frac{1}{m!} \mathcal{Y}_m(1!C_1, 2!C_2, \dots, m!C_m) = \sum_{\sum j m_j = m} \prod_{j=1}^m \frac{C_j^{m_j}}{m_j!}. \quad (16.16)$$

The first terms are

$$1 + \frac{C_1}{n} + \frac{C_1^2}{2n^2} + \frac{C_3 + C_1^3/6}{n^3} + \frac{C_1C_3 + C_1^4/24}{n^4} + \dots$$

This is an all-order Poincaré expansion with the compact-parameter uniformity stated above, not a claim of convergence of the inverse- $n$  series. Its formal  $\tau \rightarrow 0$  limit recovers ordinary binomial Stirling asymptotics:  $h_\tau \rightarrow 0$ , the rate tends to the entropy, and the prefactor tends to  $(2\pi n\alpha b)^{-1/2}$ . Endpoint regimes  $\alpha \rightarrow 0, 1$  require a different uniform analysis and are not included in the theorem.

## 17 Gaussian filters and geometric uniform convolutions

This application returns to the Fabius–Rvachev setting of the source repository. Its geometric-uniform probability model is developed in [2]. Here the point is the exact coefficient mechanism: Bell polynomials make finite-convolution moments polynomial in one geometric tail parameter, and Gaussian coefficients then remove that parameter by a finite filter.

### 17.1 Moments from finite cumulant formulae

Let  $0 < q < 1$ , let  $U_0, U_1, \dots$  be independent uniform random variables on  $[-1, 1]$ , and set

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad X_{q,N} = (1 - q) \sum_{j=0}^{N-1} q^j U_j. \quad (17.1)$$

The first series converges absolutely, even pointwise for all choices of the  $U_j$ , since its absolute value is bounded by one. Independence and  $\mathbb{E}e^{tU} = \sinh(t)/t$  give

$$\mathbb{E}e^{tX_q} = \prod_{j \geq 0} \frac{\sinh((1 - q)q^j t)}{(1 - q)q^j t}. \quad (17.2)$$

The finite products converge uniformly on compact  $t$ -sets: each factor is  $1 + O(q^{2j})$  there, and the tails of  $X_q$  also converge uniformly, justifying the expectation limit directly.

**Theorem 17.1** (Closed even moments). *Let  $\gamma_j$  be the cumulants of  $X_q$ . Then*

$$\gamma_{2r+1} = 0 \quad (r \geq 0), \quad \gamma_{2r} = \frac{2^{2r} B_{2r}}{2r} \frac{(1 - q)^{2r}}{1 - q^{2r}} \quad (r \geq 1). \quad (17.3)$$

Consequently,

$$\mathbb{E}X_q^{2d} = \mathcal{Y}_{2d}(0, \gamma_2, 0, \gamma_4, \dots, \gamma_{2d}) \quad (17.4)$$

$$= (2d)! \sum_{\sum_{r=1}^d r m_r = d} \prod_{r=1}^d \frac{1}{m_r!} \left( \frac{\gamma_{2r}}{(2r)!} \right)^{m_r}. \quad (17.5)$$

For  $X_{q,N}$  replace  $\gamma_{2r}$  by  $\gamma_{2r}(1 - q^{2rN})$ .

*Proof.* From the Bernoulli generating function,

$$\log \frac{\sinh z}{z} = \sum_{r \geq 1} \frac{2^{2r} B_{2r}}{2r(2r)!} z^{2r}.$$

For example, it follows by writing  $\sinh z/z = e^{-z}(e^{2z} - 1)/(2z)$  and applying the previously proved logarithm expansion; the linear terms cancel. Sum this identity over  $z = (1 - q)q^j t$  in a sufficiently small disk around zero. The geometric sum is  $(1 - q^{2r})^{-1}$  for the infinite model, or  $(1 - q^{2rN})/(1 - q^{2r})$  for the finite model. This proves the cumulant formula. Exponentiating the cumulant series gives the moment formula by the Bell identity. Each coefficient uses finitely many cumulants, so the final formulae have no remaining limiting or recurrence step.  $\square$

For  $q = 1/2$ , this centered geometric-uniform sum has density up on  $[-1, 1]$  under the Rvachev normalization of [2]. Thus, in particular,

$$\int_{-1}^1 x^2 \text{up}(x) dx = \frac{1}{9}, \quad \int_{-1}^1 x^4 \text{up}(x) dx = \frac{19}{675}. \quad (17.6)$$

The fourth cumulant is  $-2/225$  and the second is  $1/9$ , so the second number is  $3(1/9)^2 - 2/225$ . More generally every even moment at a rational base  $0 < q < 1$  is rational by Equation (17.5).

## 17.2 Exact geometric extrapolation and its residual

**Theorem 17.2** (Gaussian annihilation filter). *Fix  $d \geq 0$  and  $0 < s < 1$ , and define*

$$w_{d,j}(s) = \frac{(-1)^{d-j} s^{\binom{d-j+1}{2}}}{(s; s)_d} \left[ \begin{matrix} d \\ j \end{matrix} \right]_s, \quad 0 \leq j \leq d. \quad (17.7)$$

*For every polynomial  $P$  of degree at most  $d$  and every  $x_0$ ,*

$$P(0) = \sum_{j=0}^d w_{d,j}(s) P(x_0 s^j). \quad (17.8)$$

*More precisely, for each integer  $r \geq 0$ ,*

$$\sum_{j=0}^d w_{d,j}(s) s^{rj} = \begin{cases} 1, & r = 0, \\ 0, & 1 \leq r \leq d, \\ (-1)^d s^{\binom{d+1}{2}} \left[ \begin{matrix} r-1 \\ d \end{matrix} \right]_s, & r > d. \end{cases} \quad (17.9)$$

*Accordingly, for any formal series  $F(z) = \sum_{r \geq 0} c_r z^r$ ,*

$$\sum_{j=0}^d w_{d,j}(s) F(s^j z) = c_0 + (-1)^d s^{\binom{d+1}{2}} \sum_{r > d} \left[ \begin{matrix} r-1 \\ d \end{matrix} \right]_s c_r z^r. \quad (17.10)$$

*Proof.* The finite  $s$ -binomial theorem gives the weight generating polynomial

$$\sum_{j=0}^d w_{d,j}(s)x^j = \frac{\prod_{h=1}^d (x - s^h)}{(s; s)_d}. \quad (17.11)$$

Indeed the coefficient of  $x^j$  in the numerator is  $(-1)^{d-j} s^{(d-j)(d-j+1)/2} \begin{bmatrix} d \\ d-j \end{bmatrix}_s$ . Evaluate Equation (17.11) at  $x = 1$  and at  $x = s^r$ . For  $1 \leq r \leq d$  one factor is zero. For  $r > d$ , factor  $-s^h$  from every numerator factor to obtain

$$\frac{\prod_{h=1}^d (s^r - s^h)}{(s; s)_d} = (-1)^d s^{d(d+1)/2} \frac{(s^{r-d}; s)_d}{(s; s)_d} = (-1)^d s^{d(d+1)/2} \begin{bmatrix} r-1 \\ d \end{bmatrix}_s.$$

This proves Equation (17.9). Apply it to the monomials of  $P$  or, coefficientwise, of  $F$  to prove the remaining claims. The  $d = 0$  case is included by the empty-product convention.  $\square$

One can also derive the weights by Lagrange interpolation of  $P$  at  $x_0, x_0 s, \dots, x_0 s^d$ , when  $x_0 \neq 0$ . The polynomial generating function proof has the advantage that it simultaneously gives the exact residual on every higher monomial. For analytic  $F$ , the last identity holds wherever its power series converges at the finitely many evaluation points; no summability convention is needed.

**Corollary 17.3** (Exact finite recovery of geometric-convolution moments). *For  $d, N \geq 0$ , put  $s = q^2$ . Then*

$$\mathbb{E}X_q^{2d} = \sum_{j=0}^d w_{d,j}(q^2) \mathbb{E}X_{q,N+j}^{2d}. \quad (17.12)$$

Here  $X_{q,0} = 0$  and the zeroth moment is one, including at zero.

*Proof.* In Equation (17.5), the finite- $N$  replacement is  $\gamma_{2r} \mapsto \gamma_{2r}(1 - y^r)$  with  $y = q^{2N}$ . A profile with  $\sum r m_r = d$  therefore has degree at most  $d$  in  $y$ . Thus  $\mathbb{E}X_{q,N}^{2d} = P_d(s^N)$  for a polynomial  $P_d$  of degree at most  $d$ , whose constant term is  $\mathbb{E}X_q^{2d}$ . Apply Equation (17.8) with  $x_0 = s^N$ .  $\square$

For  $d = 1$  the filter is  $P(0) = (P(sx_0) - sP(x_0))/(1 - s)$ . The higher filters remove every geometric error  $s^{rN}$  with  $1 \leq r \leq d$  at once. This exactness has been proved for the moment sequence through its polynomial tail parameter; it does not assert exactness for arbitrary pointwise samples of a density or for a general asymptotic expansion. That qualification is essential in applications to spline approximations and Fabius values.

## 18 Verification, implementation, and further extensions

### 18.1 What was checked

The accompanying `verify_identities.py` is a reproducible test suite, not a proof assistant. Its core uses only Python's standard library: integers, rational numbers, finite permutations, and elementary polynomial arithmetic. It performs **10,267 exact checks**, all of which passed in the run recorded in `verification_results.txt`. The checks include small and boundary indices, bases  $q = 1/2, 2/3, 2$ , polynomial congruences modulo cyclotomic polynomials, and the specializations  $q = 1, -1$ . The per-family counts are recorded in the log rather than hidden behind a single success message.

The comparisons deliberately use different constructions. Gaussian products are compared with polynomial Pascal recurrences and subset enumeration;  $q$ -Stirling finite sums with node recurrences;  $q$ -Eulerian coefficients with direct permutation enumeration; Jackson chain rules with barycentric divided differences; inverse coefficients with independent series substitution; and moments from cumulants with direct products of elementary uniform moment-generating series. Gaussian Lucas

evaluations are tested as polynomial congruences, not by floating-point substitutions near roots of unity. The local derivative tests at  $q = -1$  first remove the zero factor.

Finite testing cannot establish a theorem for arbitrary indices. The proofs above carry that burden. Conversely, a test failure would expose at least a transcription, convention, or implementation error even if a nearby theorem were valid. These complementary roles are particularly valuable for the powers of  $q$  in inversion and for the distinction between  $S_q(n, k)$  and  $q^{\binom{k}{2}} S_q(n, k)$ .

## 18.2 A numerical check of the double-scaling order

An optional final experiment uses `mpmath` with 70 decimal digits. For  $\tau = 1.3$  and  $\alpha = 1/3$ , let  $L_0(n)$  be the leading line of Equation (16.10), and define

$$R_1(n) = \log \left[ \frac{n}{n/3} \right]_{e^{-1.3/n}} - L_0(n) - C_1/n, \quad R_2(n) = R_1(n) - C_3/n^3.$$

The actual output, rounded here, is

| $n$             | $n^3 R_1(n)$               | $n^5 R_2(n)$                |
|-----------------|----------------------------|-----------------------------|
| 60              | 0.0815396566910922         | -0.1977273873208401         |
| 120             | 0.0815808310074465         | -0.1979993937823302         |
| 240             | 0.0815911422914091         | -0.1980676188797450         |
| Predicted limit | $C_3 = 0.0815945809653480$ | $C_5 = -0.1980903805565352$ |

The approach to the two predicted constants checks the signs and orders in the formulae. These are high-precision numerical observations, not interval-certified inequalities. The uniform error estimate is proved independently in Section 16.

## 18.3 How to use the closed formulae

For a single coefficient, partition-profile formulae avoid generating unneeded lower coefficients. For a whole triangular array, a recurrence or truncated-series multiplication is often faster, even when a closed formula is available. “Closed” and “computationally cheapest” are different properties. In particular, alternating Gaussian sums can lose precision if evaluated numerically near  $q = 1$ ; use polynomial arithmetic, the Bernoulli expansion of Section 14, or sufficient exact precision instead. At roots of unity, use the polynomial, cyclotomic, or Lucas form before evaluating a rational quotient.

The spectral Newton formula is useful when the nodes are distinct and few, while the homogeneous-symmetric formula remains regular at coalescing nodes. The Bell formula is most effective when logarithmic coefficients are simple. The Gaussian filter is exact only after the finite-degree dependence on the tail parameter has been established. These selection rules follow from the proofs, rather than from a preference for one notation.

## 18.4 Directions not claimed as results here

The framework suggests several further investigations. First, one can replace one geometric alphabet by several alphabets with different bases. The Bell-product theorem continues immediately after replacing  $[N]_{q^j}$  by the corresponding sum of geometric power sums, but a single Gaussian connection array need no longer exist. A useful problem is to classify which node changes preserve particularly simple inverse matrices.

Second, ordinary multivariate Lagrange inversion can accept the same finite  $q$ -product inputs. The determinant and Jacobian factors remain ordinary; only their coefficients acquire Gaussian or geometric-power structure. Working out a systematic multivariate analogue of Equation (12.4) would extend the parent article’s coefficient calculus without conflating it with noncommutative substitution.



Third, the regimes  $q \rightarrow \zeta$  with  $n \rightarrow \infty$ , or  $q = e^{-\tau/n}$  with  $k/n \rightarrow 0$ , lie outside the uniform theorem proved here. The local cyclotomic decomposition and the separate fixed- $k$  formula provide starting points for matched expansions, but no uniform transition theorem for those regimes is asserted. In particular, the present inverse- $n$  expansion is not advertised as a complete exponentially improved transseries.

## 19 Conclusion

The main organizing observation is that Gaussian coefficients are Newton connection coefficients for geometric nodes. It explains why  $q$ -binomial inversion,  $q$ -Stirling sums, colored-node families, and geometric divided differences fit the same triangular-array calculus. Weighted permutation sorting then joins that basis calculus to  $q$ -Eulerian polynomials. On the other side, logarithms of finite  $q$ -products have geometric power-sum coefficients, so the parent's ordinary Bell engine applies without alteration. It produces powers, derivatives, local jets, inverse coefficients, cumulants, and asymptotic corrections in one uniform language.

The places where the language must change have been made explicit. Jackson differentiation needs geometric shifts; its chain rule is a path sum of divided differences. Ordinary substitution keeps the ordinary Lagrange factor  $r/n$ . A  $q$ -factorial normalization alone need not give polynomial  $q$ -weights. Root-of-unity evaluation must be performed after cancellation, and fixed-base asymptotics must be separated from a moving-base limit. With these distinctions in place,  $q$ -binomial coefficients become a particularly effective extension of combinatorial coefficient calculus rather than a source of ambiguous formal analogies.

## A A formula-selection guide

The following guide lists the finite mechanisms most useful for exact calculations. Each entry points to a formula with all indices and normalizations specified in the main text.

| Task                                                     | Formula and applicable setting                                                        |
|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Evaluate a Gaussian polynomial                           | Subset or rectangle sums (2.6)–(2.7); no division, hence valid at every complex $q$ . |
| Convolve or invert a Gaussian transform                  | Vandermonde (3.3), multiblock (3.4), inversion (3.7).                                 |
| Change from geometric to affine-geometric nodes          | The finite mixed ordinary/Gaussian sum (5.14).                                        |
| Compute $q$ -Stirling entries without recurrence         | (6.4), (6.5), (6.7); polynomial cancellation at exceptional bases.                    |
| Convert Stirling and Eulerian data                       | (7.5) and (7.6), using the major-index convention (7.1).                              |
| Extract a coefficient of mixed $q$ -products             | The Bell or finite profile formulae (8.4)–(8.5).                                      |
| Differentiate on a geometric grid                        | Finite Jackson sum (9.5); product (9.6); path chain rule (10.1).                      |
| Compose $q$ -normalized ordinary series                  | (11.5) and (11.7); the partial coefficients can be rational in $q$ .                  |
| Invert an ordinary series with finite $q$ -product input | (12.3) or (12.4); the prefactor is ordinary $r/n$ .                                   |
| Compute derivatives at $q = 1$                           | Bernoulli cumulants (14.4) and the finite Stirling–Bell sum (14.6).                   |
| Evaluate or expand at a root of unity                    | Lucas (15.3); regularized local jet (15.6).                                           |
| Extract moving-base asymptotic coefficients              | (16.11) and (16.16), under the compactness assumptions (16.6).                        |
| Remove a finite geometric tail polynomial                | Gaussian weights (17.7), exact residual (17.10), moment application (17.12).          |

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